

Genome Sequencing

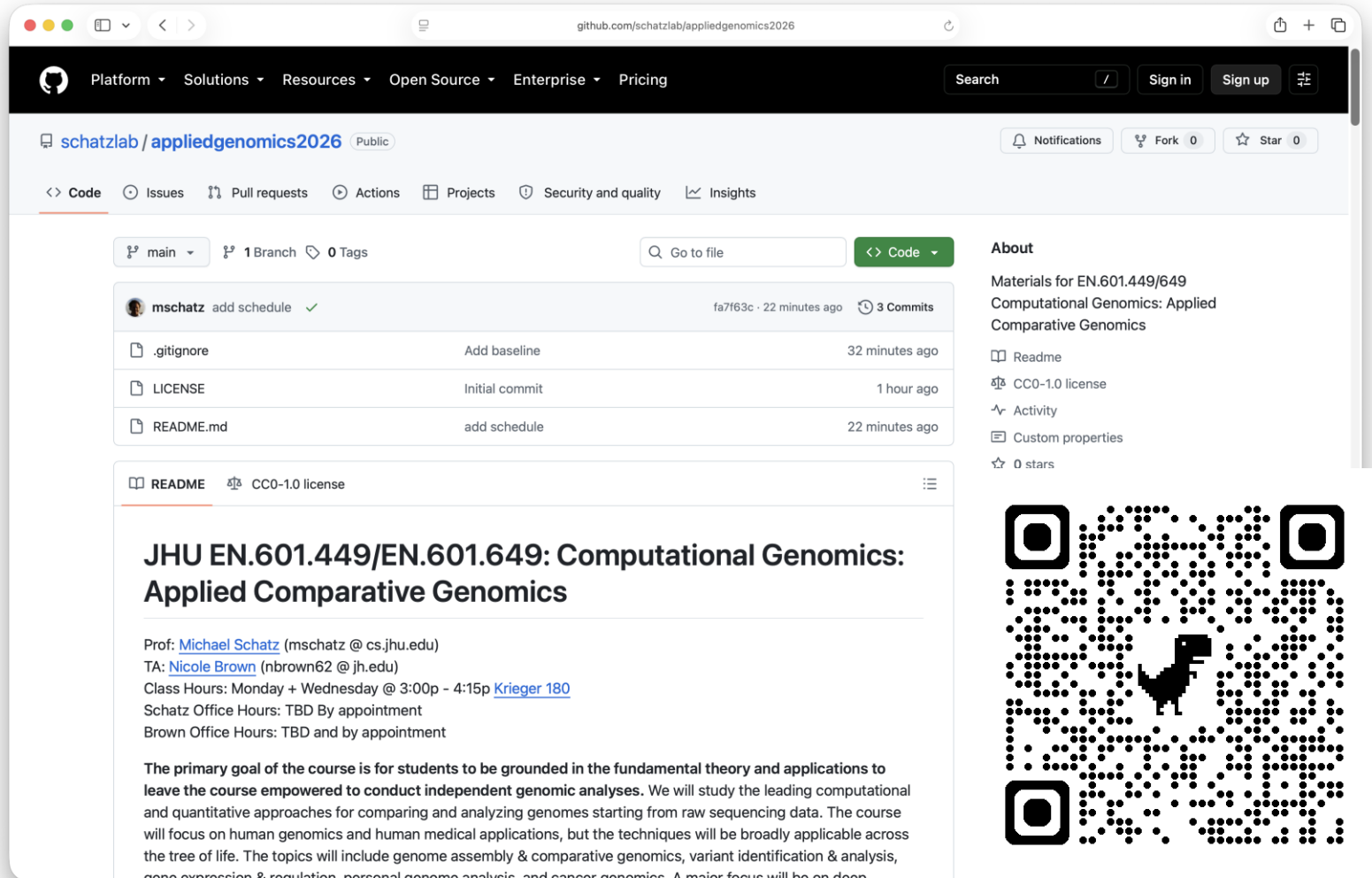
Michael Schatz & Nicole Brown

Sept 2 2026

Lecture 2: Applied Comparative Genomics



Course Webpage



github.com/schatzlab/appliedgenomics2026

Platform Solutions Resources Open Source Enterprise Pricing

Search Sign in Sign up

schatzlab / appliedgenomics2026 Public

Notifications Fork 0 Star 0

<> Code Issues Pull requests Actions Projects Security and quality Insights

main 1 Branch 0 Tags

Go to file Code

mschatz add schedule ✓ fa7f63c · 22 minutes ago 3 Commits

.gitignore	Add baseline	32 minutes ago
LICENSE	Initial commit	1 hour ago
README.md	add schedule	22 minutes ago

README CC0-1.0 license

JHU EN.601.449/EN.601.649: Computational Genomics: Applied Comparative Genomics

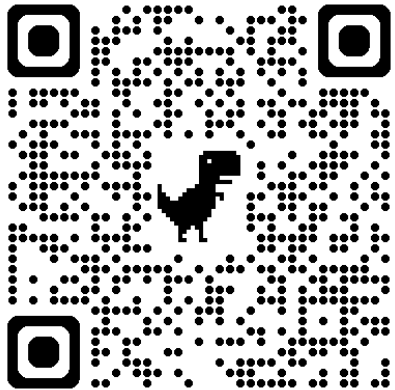
Prof: [Michael Schatz](#) (mschatz @ cs.jhu.edu)
TA: [Nicole Brown](#) (nbrown62 @ jh.edu)
Class Hours: Monday + Wednesday @ 3:00p - 4:15p [Krieger 180](#)
Schatz Office Hours: TBD By appointment
Brown Office Hours: TBD and by appointment

The primary goal of the course is for students to be grounded in the fundamental theory and applications to leave the course empowered to conduct independent genomic analyses. We will study the leading computational and quantitative approaches for comparing and analyzing genomes starting from raw sequencing data. The course will focus on human genomics and human medical applications, but the techniques will be broadly applicable across the tree of life. The topics will include genome assembly & comparative genomics, variant identification & analysis, gene expression & regulation, personal genome analysis, and cancer genomics. A major focus will be on deep

About

Materials for EN.601.449/649
Computational Genomics: Applied
Comparative Genomics

Readme
CC0-1.0 license
Activity
Custom properties
0 stars



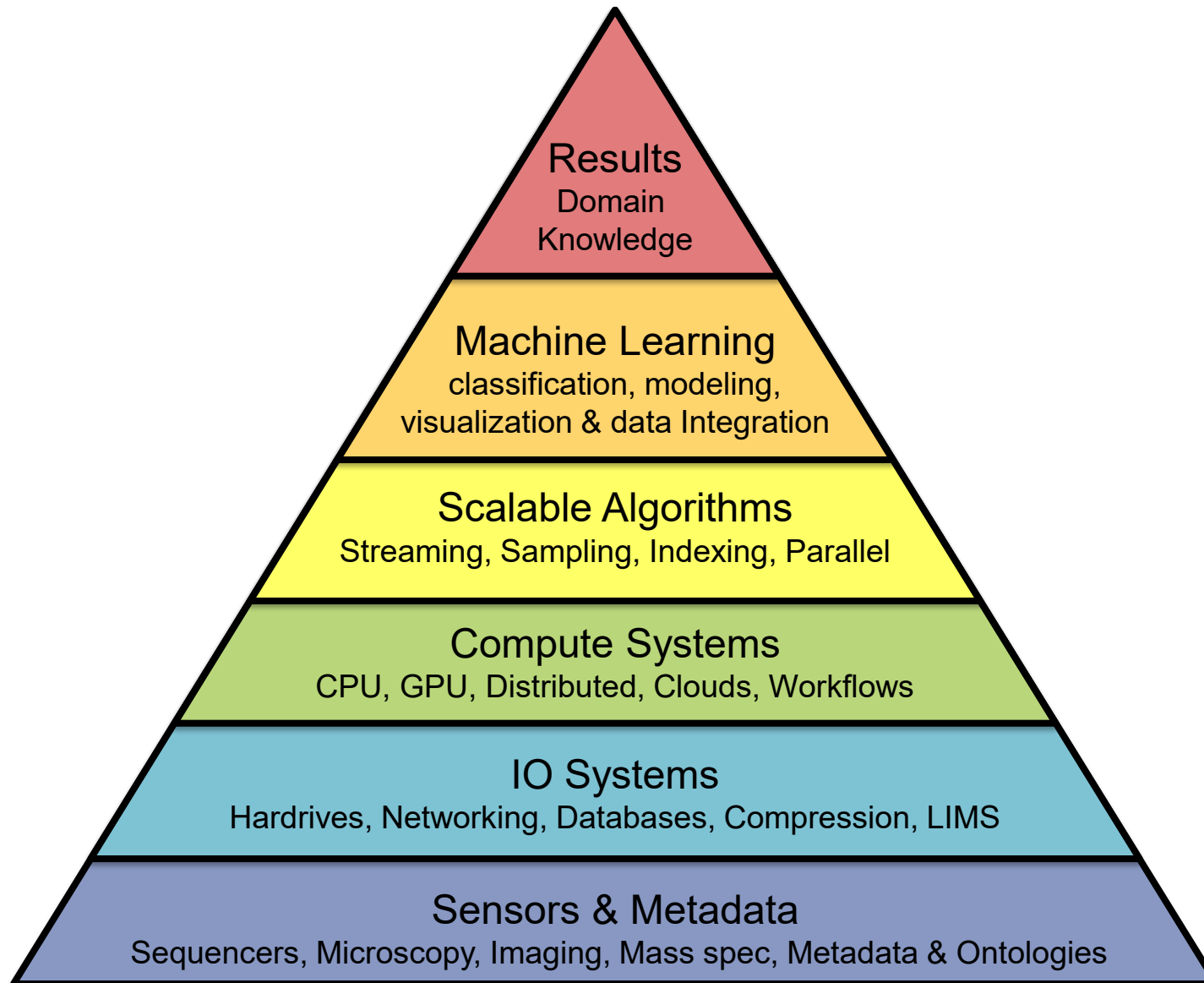
<https://github.com/schatzlab/appliedgenomics2026>

TA: Nicole Brown

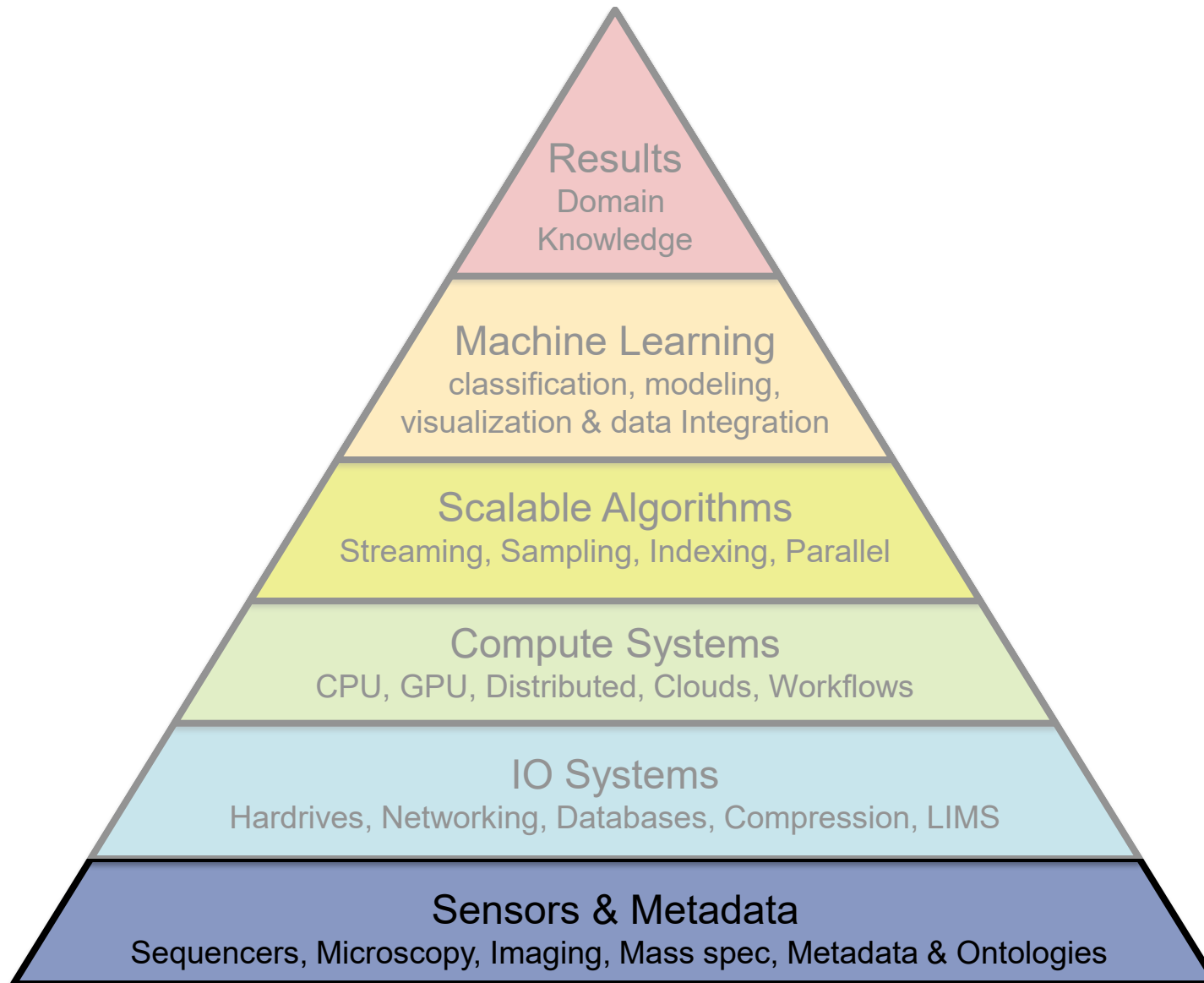


Office Hours: Wednesdays
@ 4:30pm, Malone 216 and Zoom

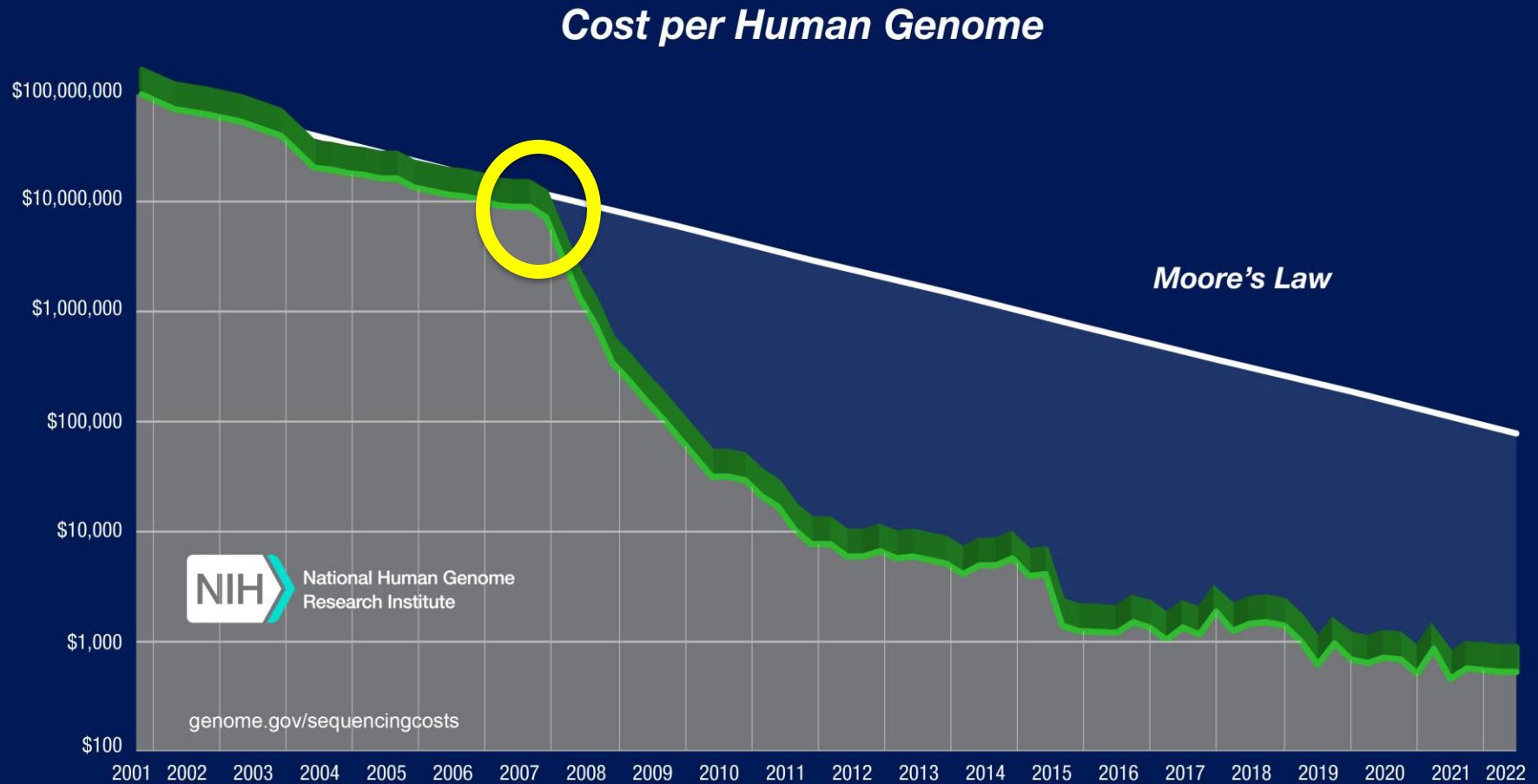
Comparative Genomics Technologies



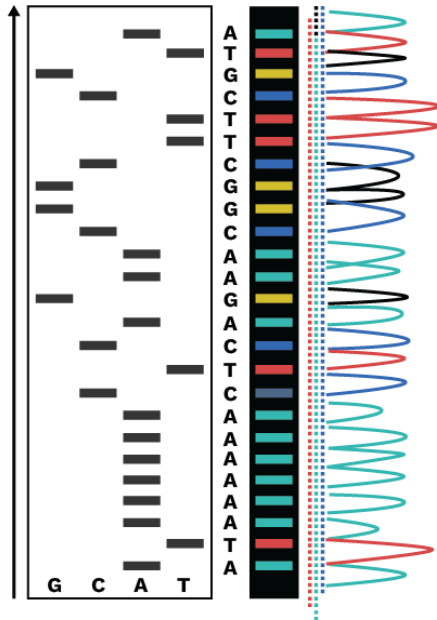
Comparative Genomics Technologies



Cost per Genome



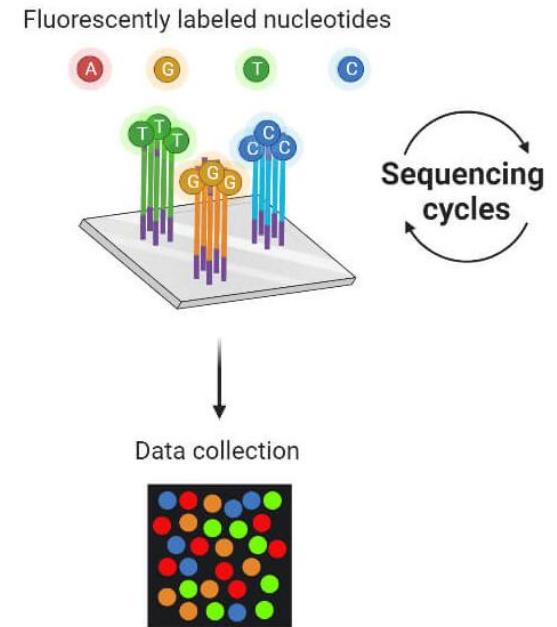
Second Generation Sequencing



Sanger Sequencing

Slow, expensive

One fragment at a time

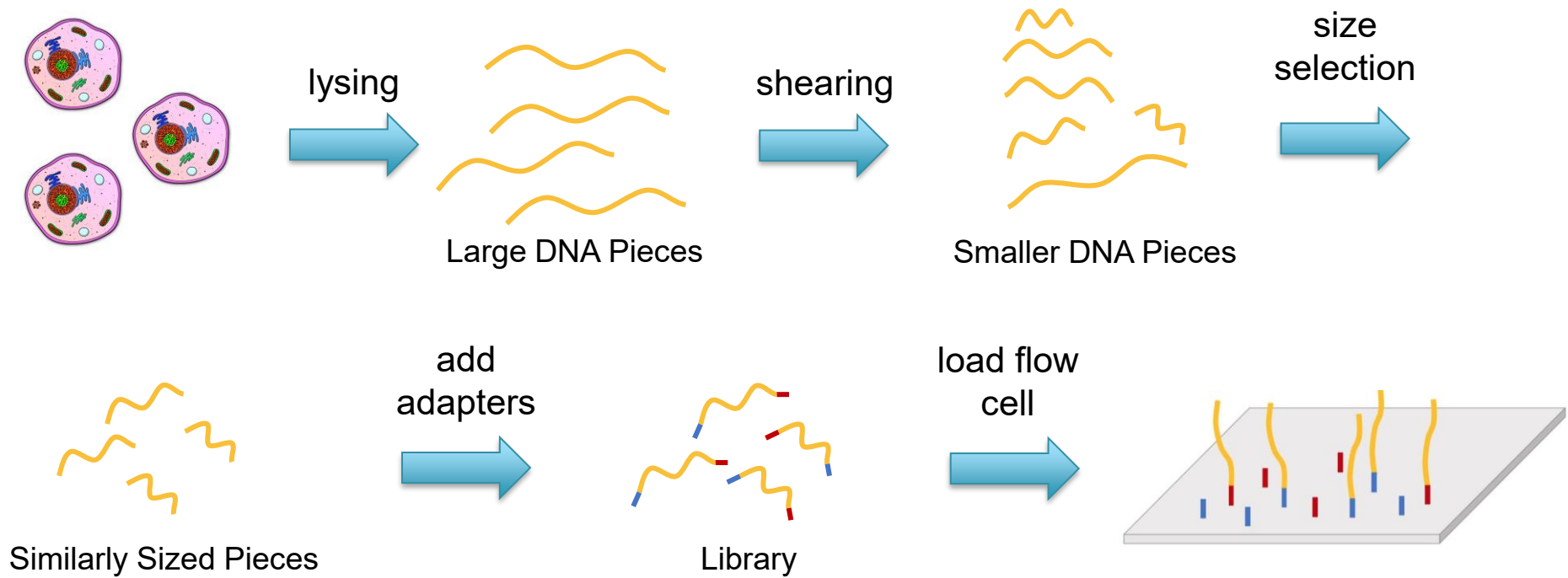


Illumina's Sequencing by
Synthesis

Fast, cheaper

Many fragments at once

Illumina Library Prep

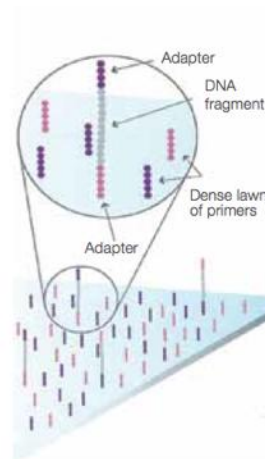


Second Generation Sequencing

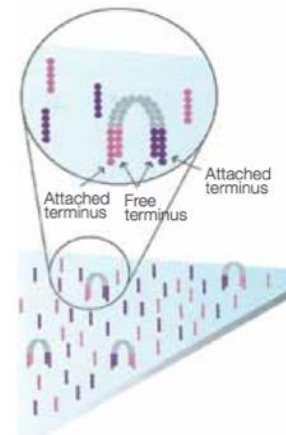


Illumina NovaSeq 6000
Sequencing by Synthesis

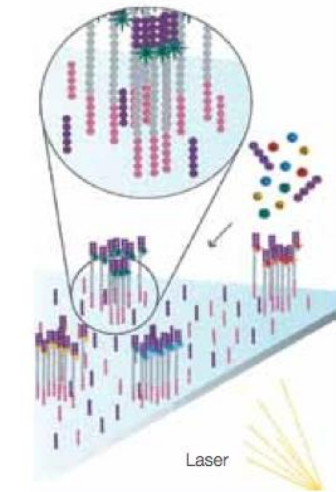
>3Tbp / day
(JHU has 4 of these!)



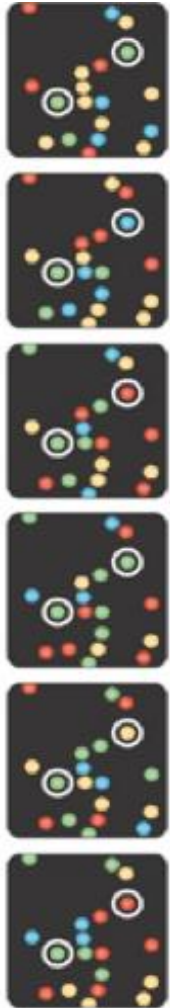
1. Attach



2. Amplify

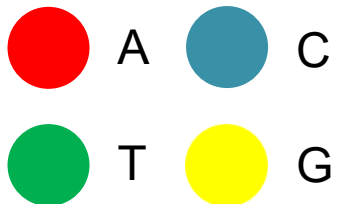
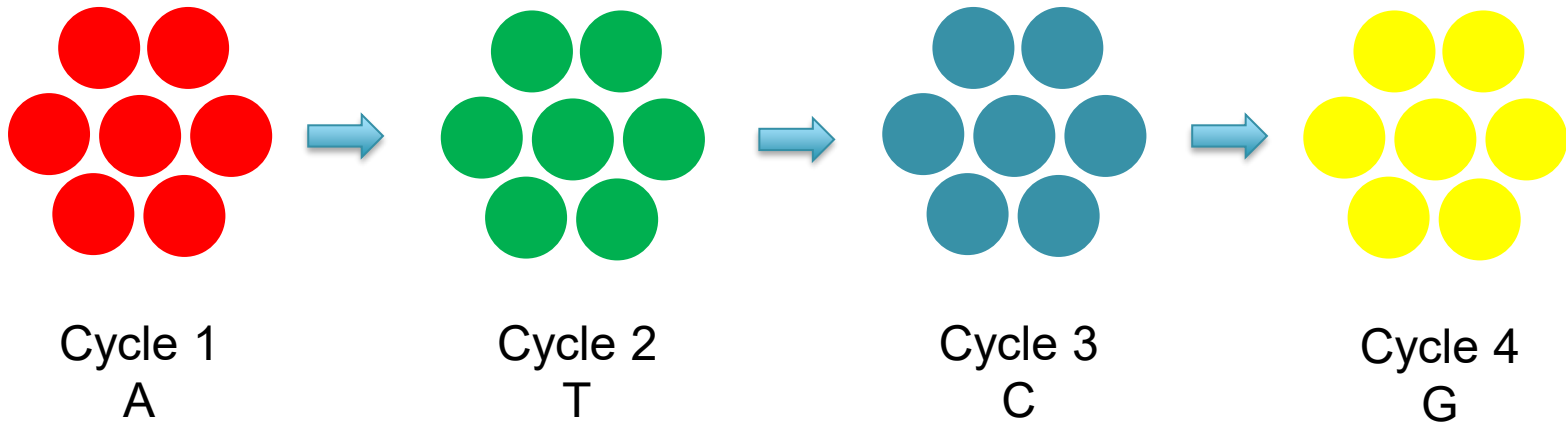


3. Image

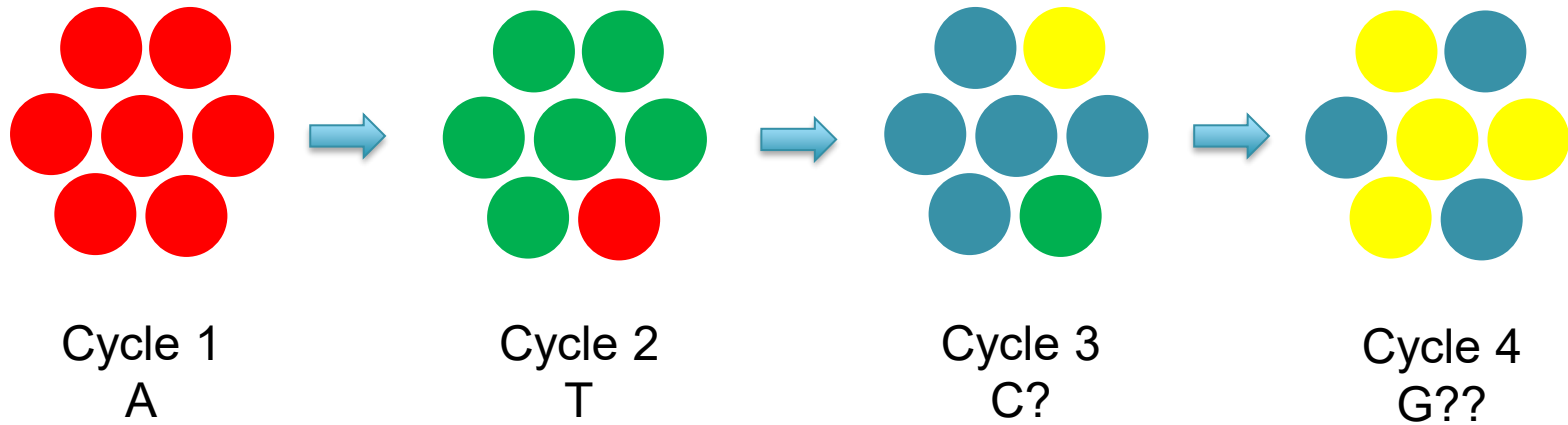


Metzker (2010) Nature Reviews Genetics 11:31-46
<https://www.youtube.com/watch?v=fCd6B5HRaZ8>

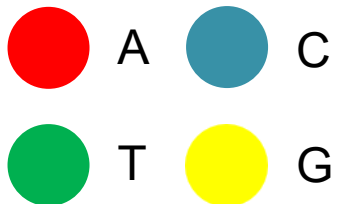
Cluster Basecalling



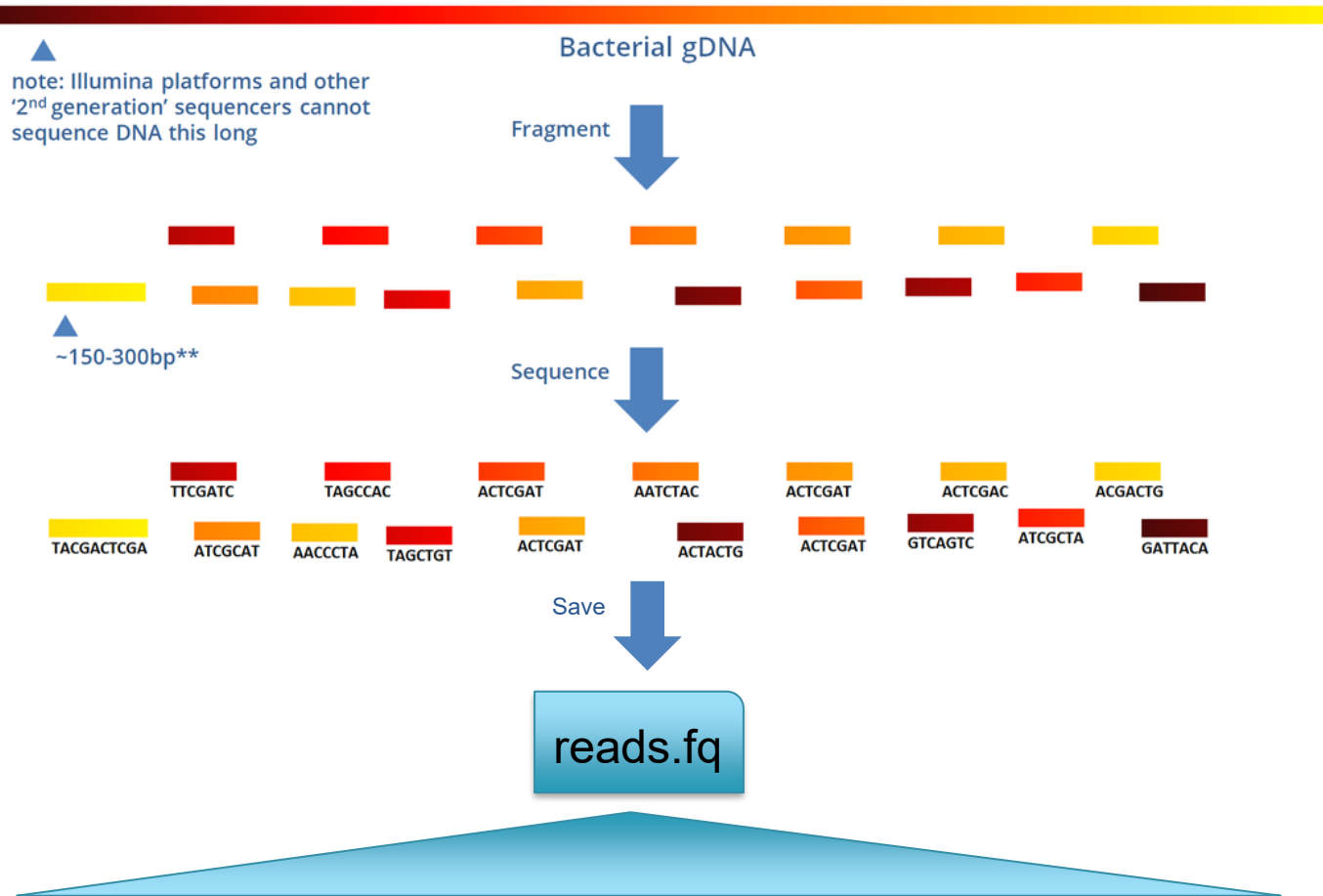
Cluster Basecalling



The sequencer isn't always confident in a base's identity



FASTQ Files



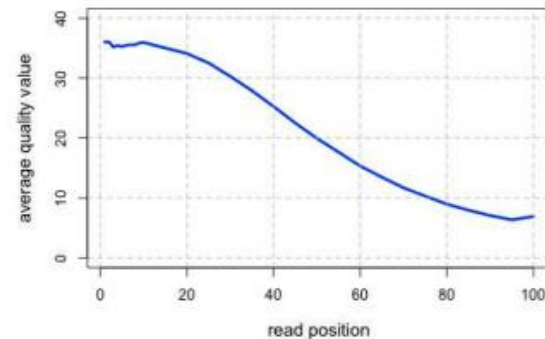
```
@SEQ_ID
GATTTGGGGTTCAAAGCAGTATCGATCAAATAGTAAATCCATTTGTTCAACTCACAGTTT
+
!''*((( (**+))%%#+)(%%%).1***-+*''))**55CCF>>>>>CCCCCCC65
```

@Identifier
Sequence
+Separator
Quality Values
...

Illumina Quality

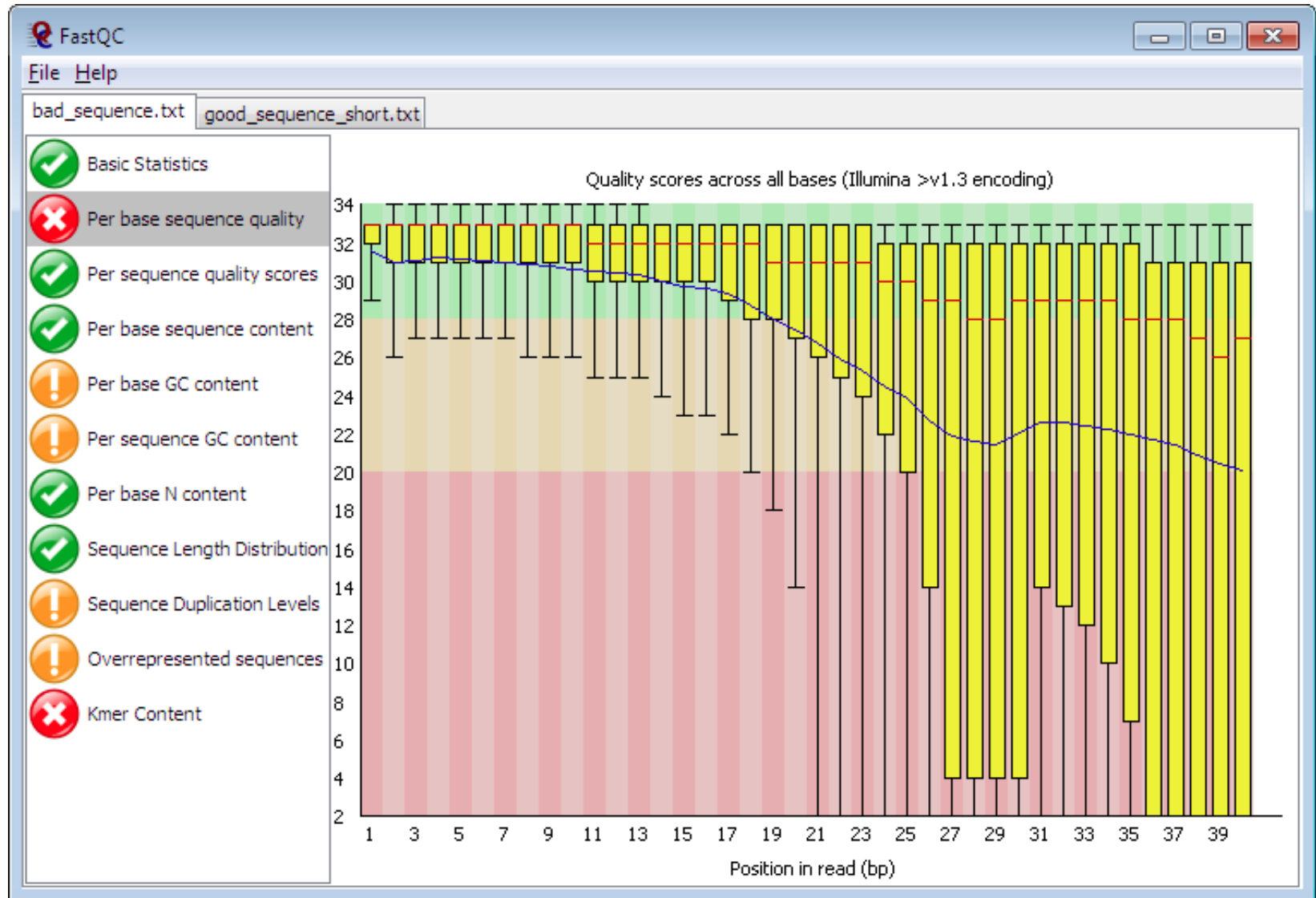
QV	P _{error}
40	1/10000
30	1/1000
20	1/100
10	1/10

$$Q_{\text{sanger}} = -10 \log_{10} p$$

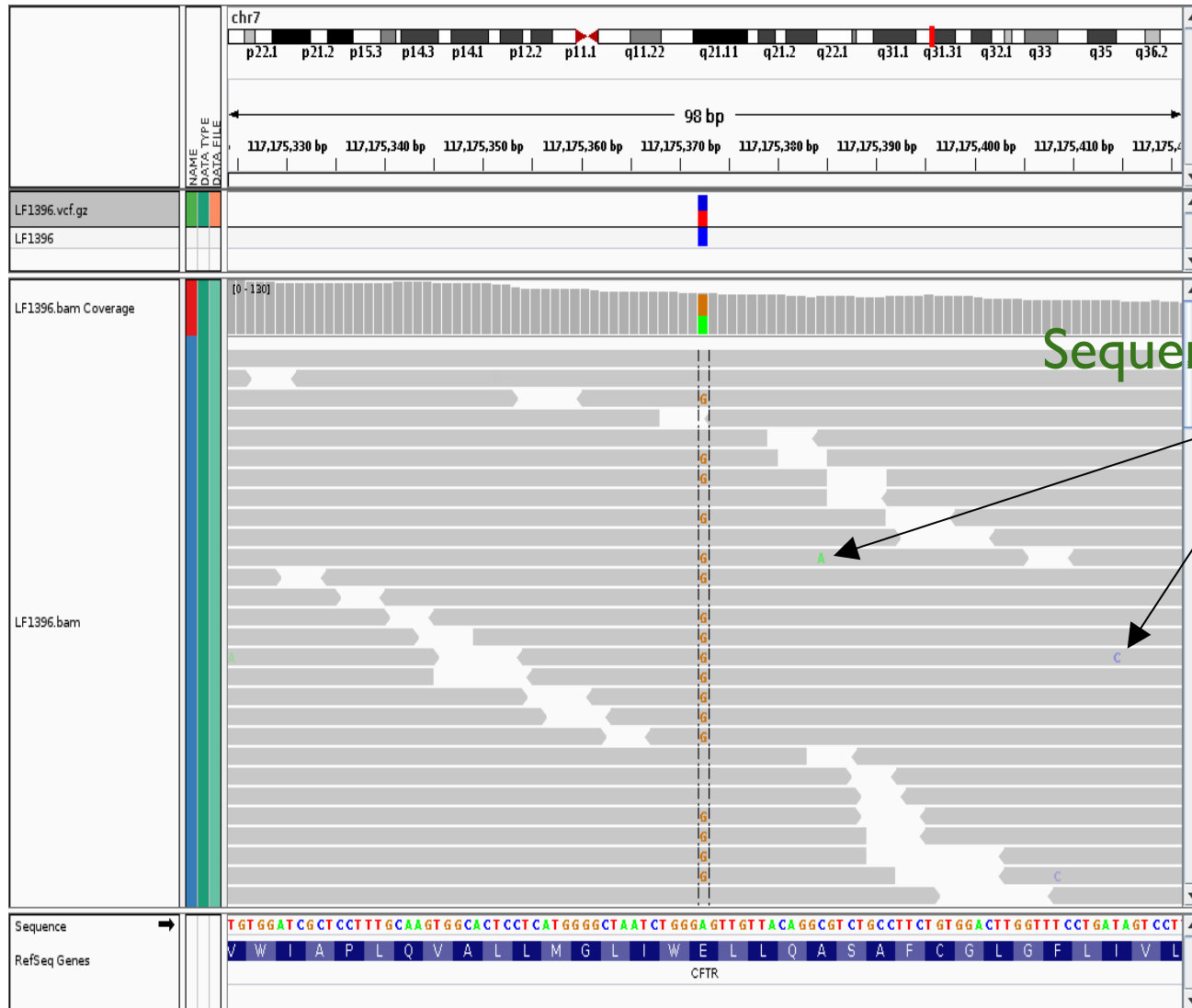
[illegible]

- ```
S - Sanger Phred+33, raw reads typically (0, 40)
X - Solexa Solexa+64, raw reads typically (-5, 40)
I - Illumina 1.3+ Phred+64, raw reads typically (0, 40)
J - Illumina 1.5+ Phred+64, raw reads typically (3, 40)
 with 0=unused, 1=unused, 2=Read Segment Quality Control Indicator (bold)
 (Note: See discussion above).
L - Illumina 1.8+ Phred+33, raw reads typically (0, 41)
```

# FASTQC: Is my data any good?



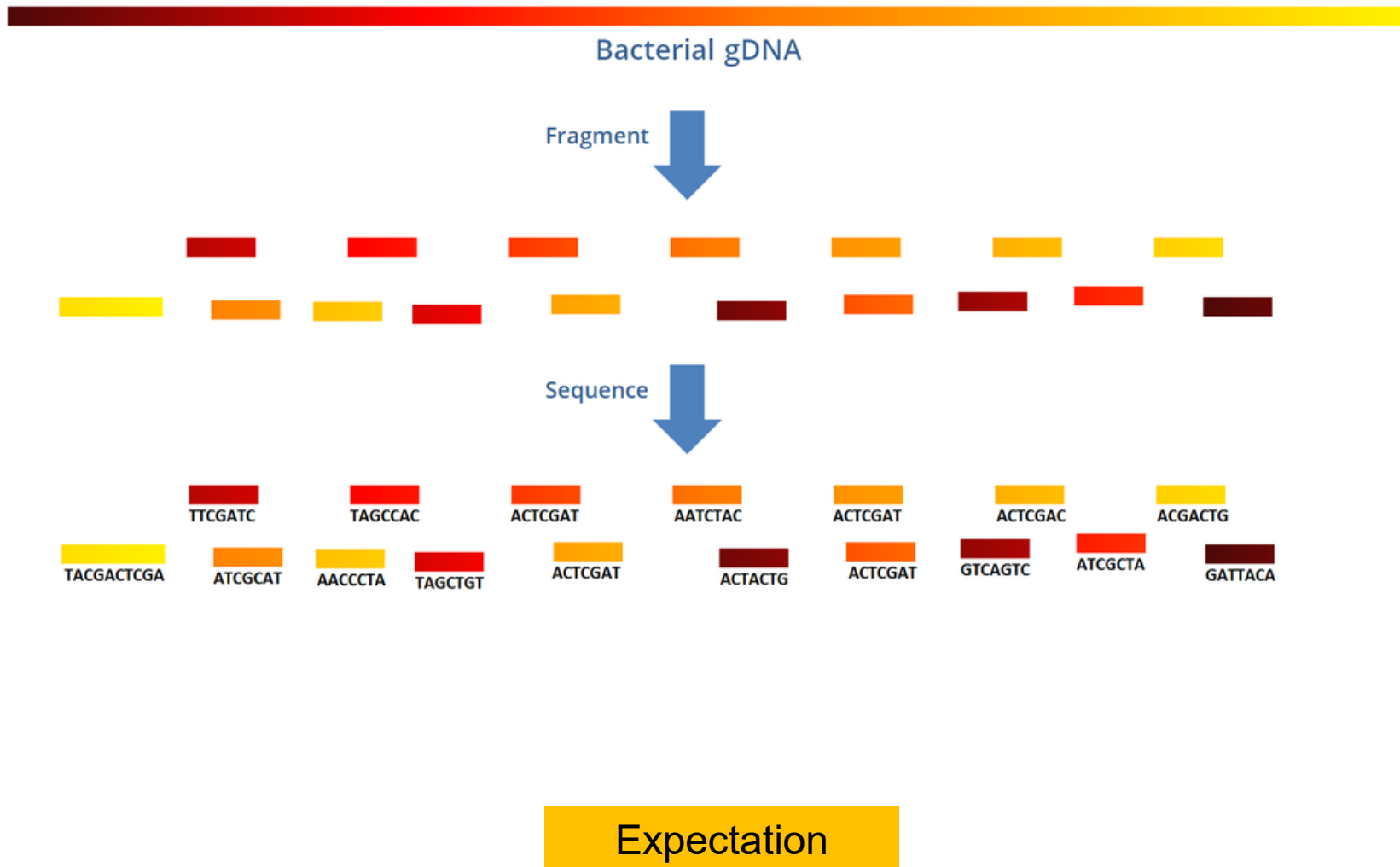
# Sequencing errors fall out as noise (most of the time)



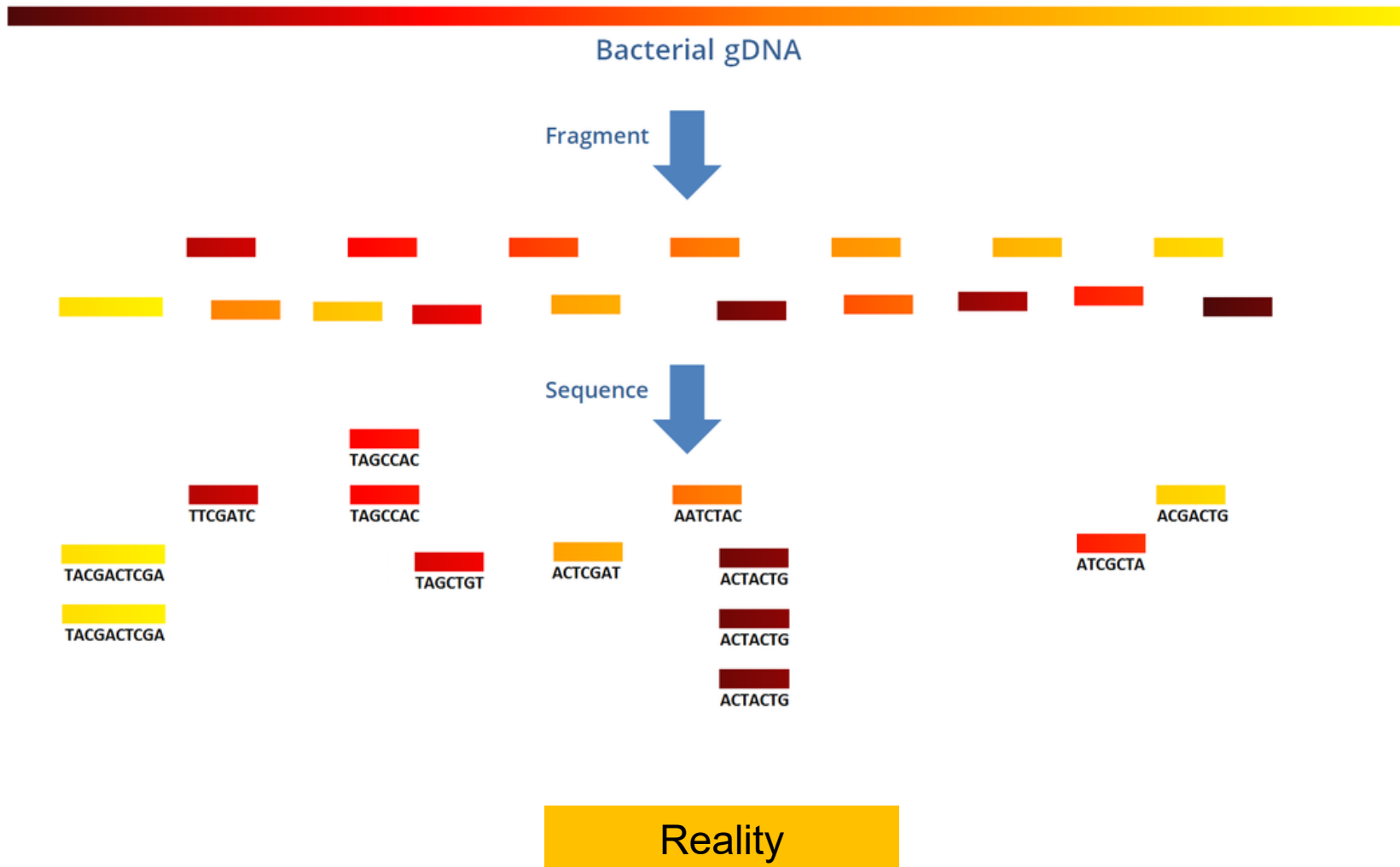
Sequencing errors



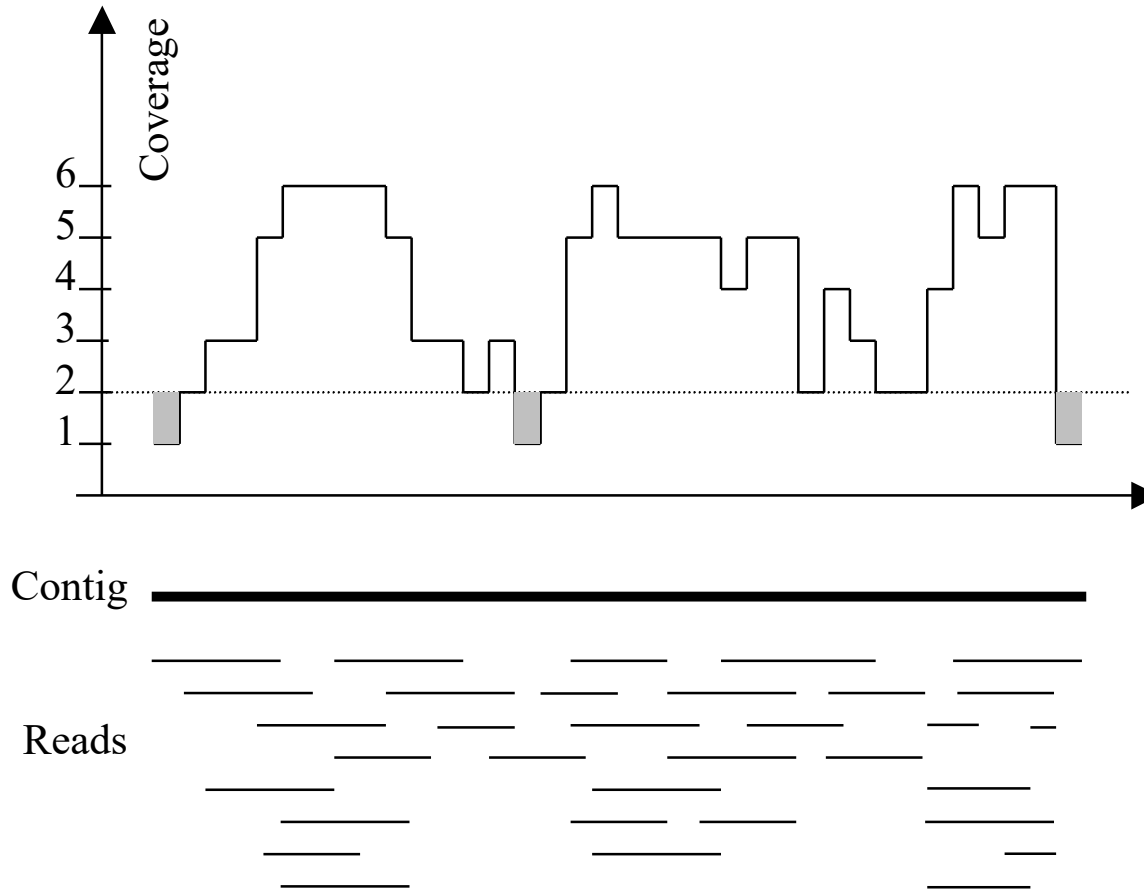
# Sequencing coverage



# Sequencing coverage



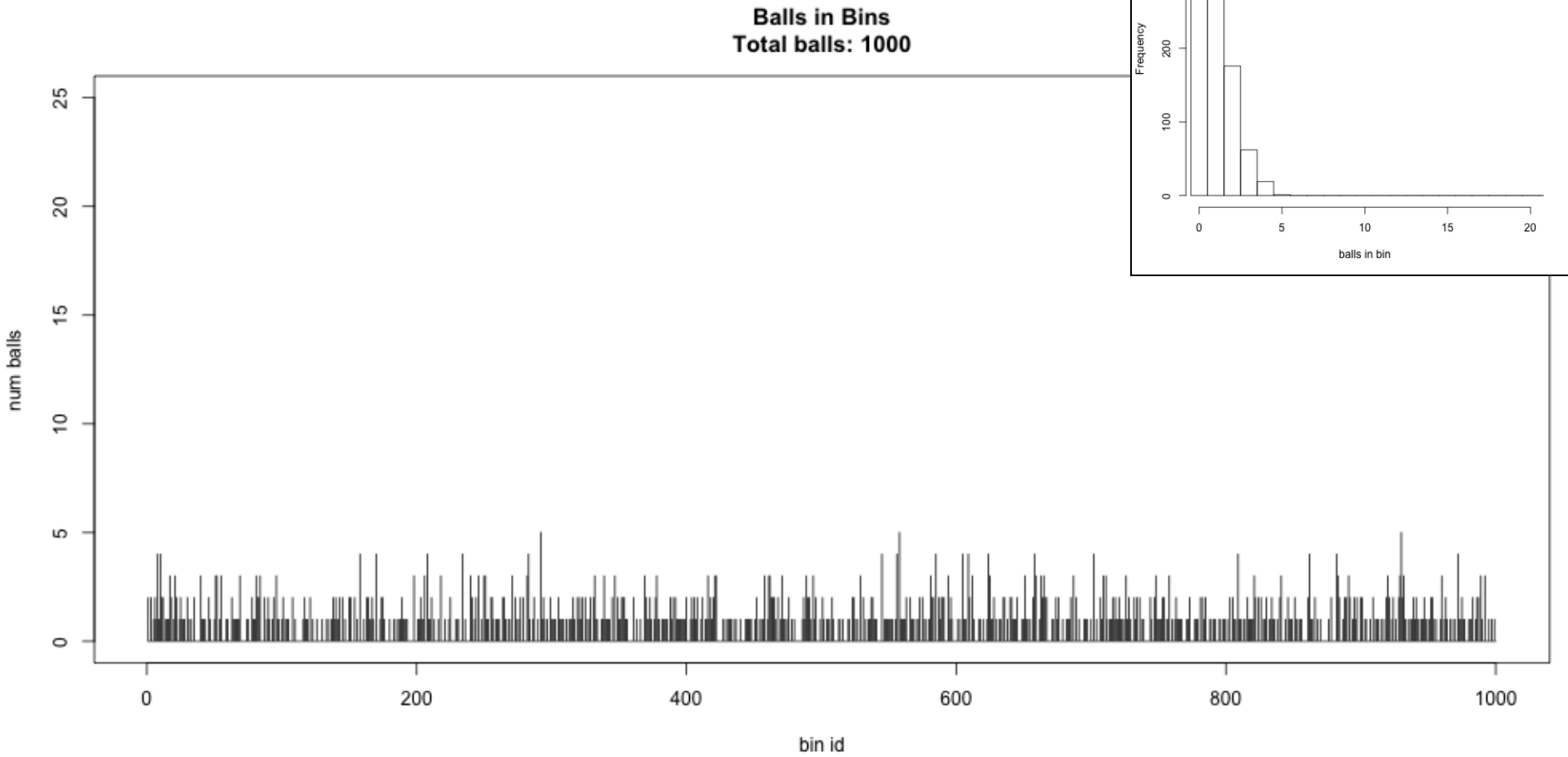
# Typical sequencing coverage



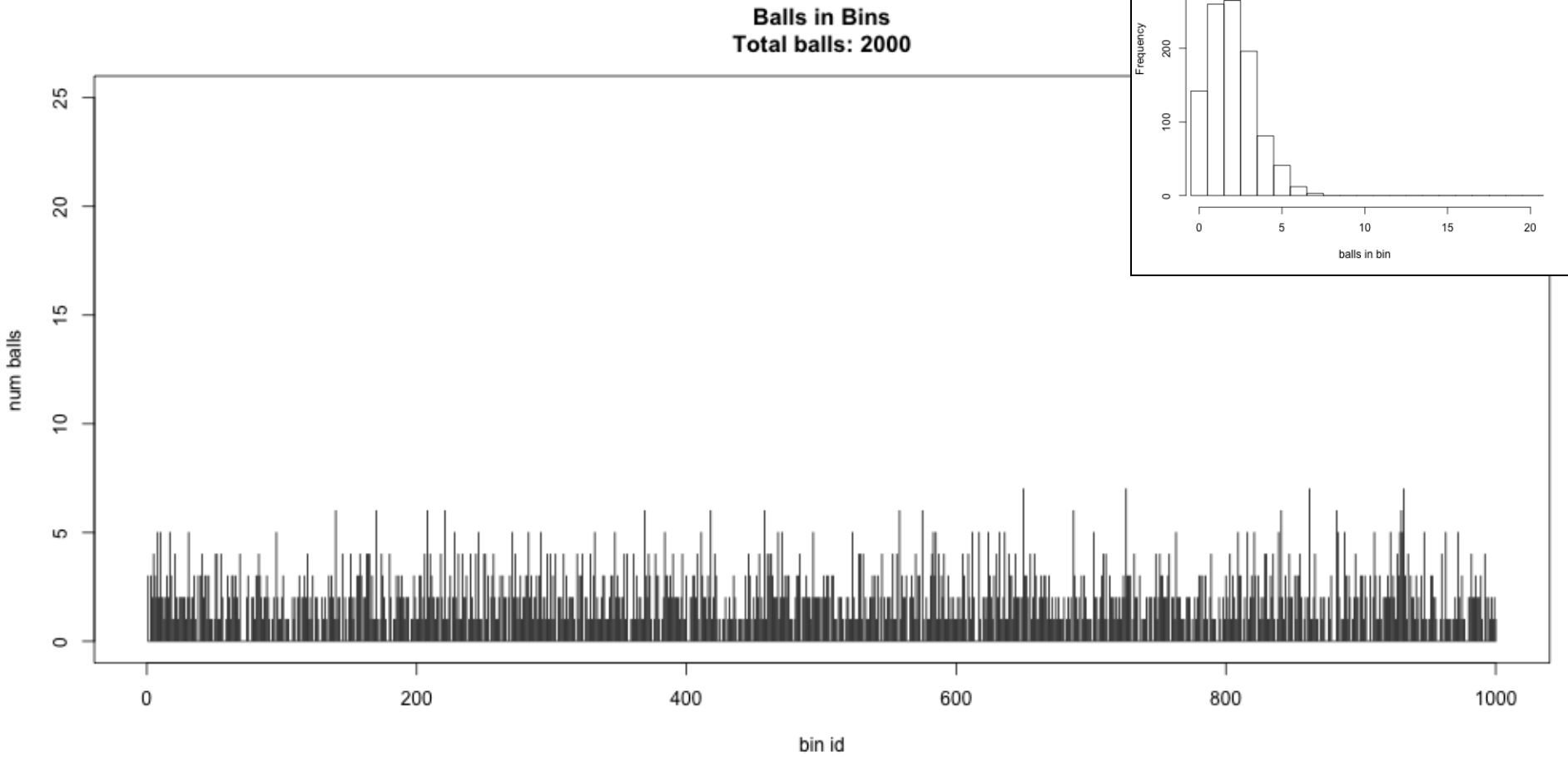
# Imagine raindrops on a sidewalk

We want to cover the entire sidewalk but each drop costs \$1

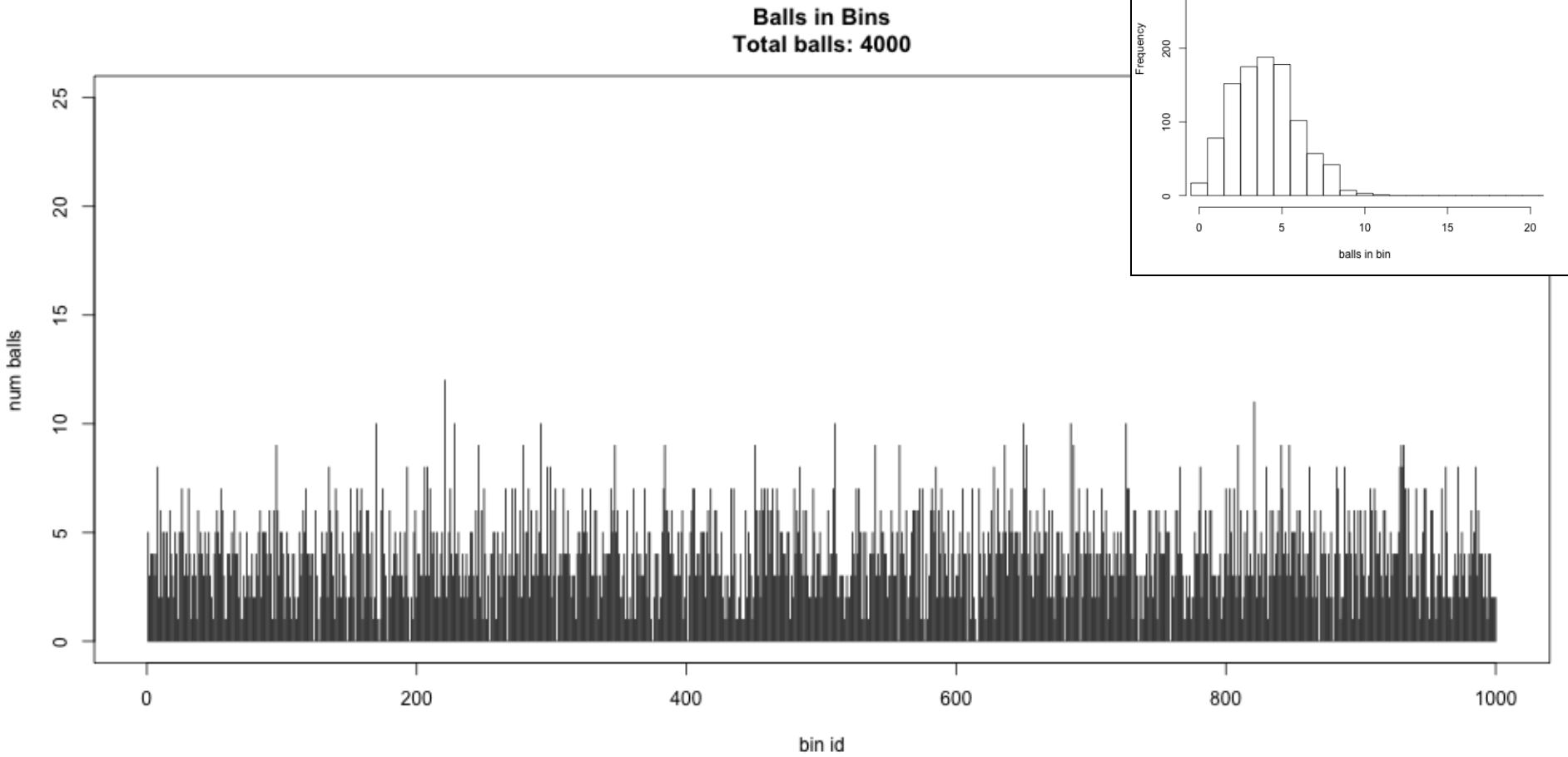
# Ix sequencing



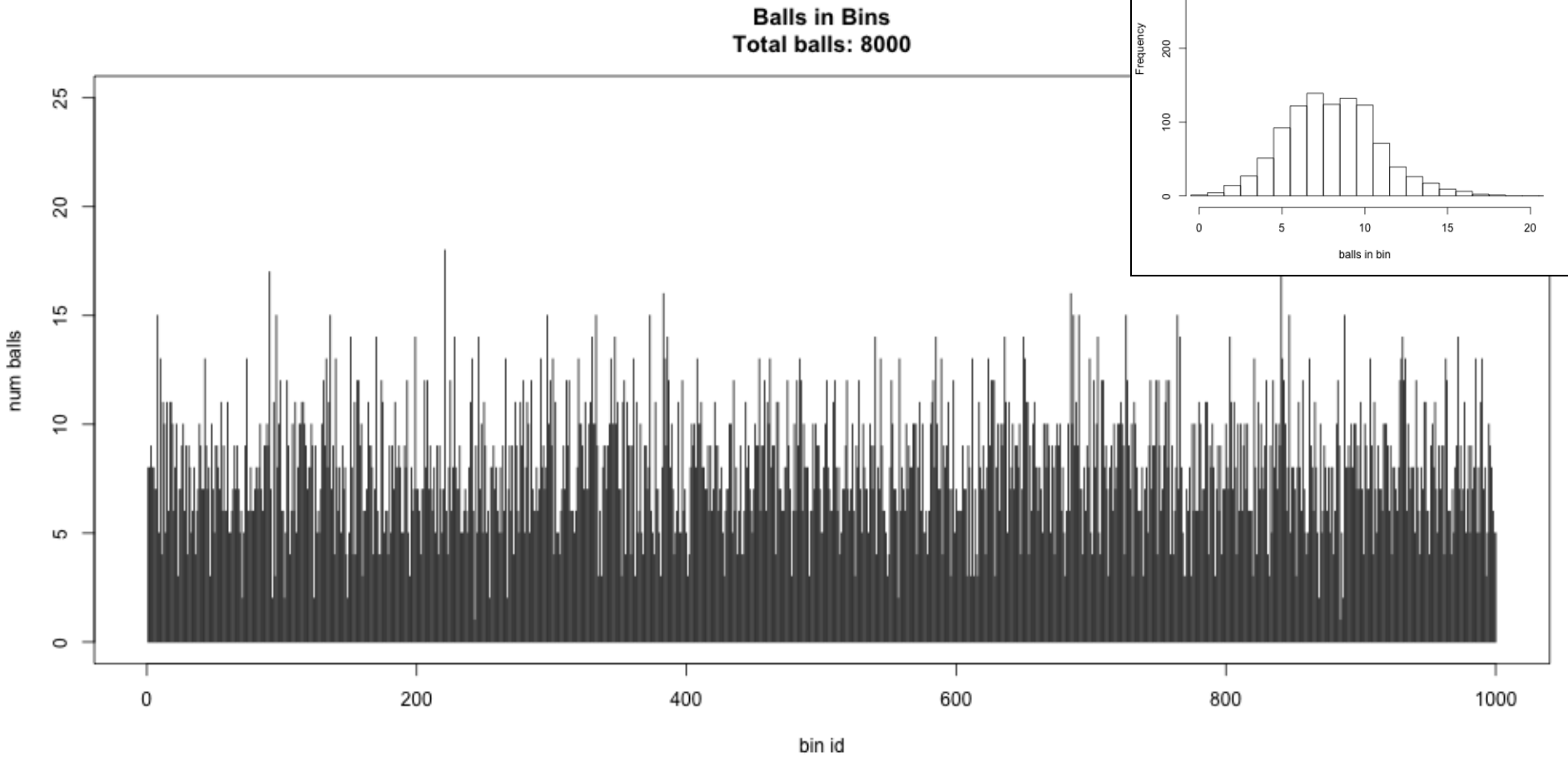
# 2x sequencing



# 4x sequencing



# 8x sequencing



# Poisson Distribution

The probability of a given number of events occurring in a fixed interval of time and/or space if these events occur with a known average rate and independently of the time since the last event.

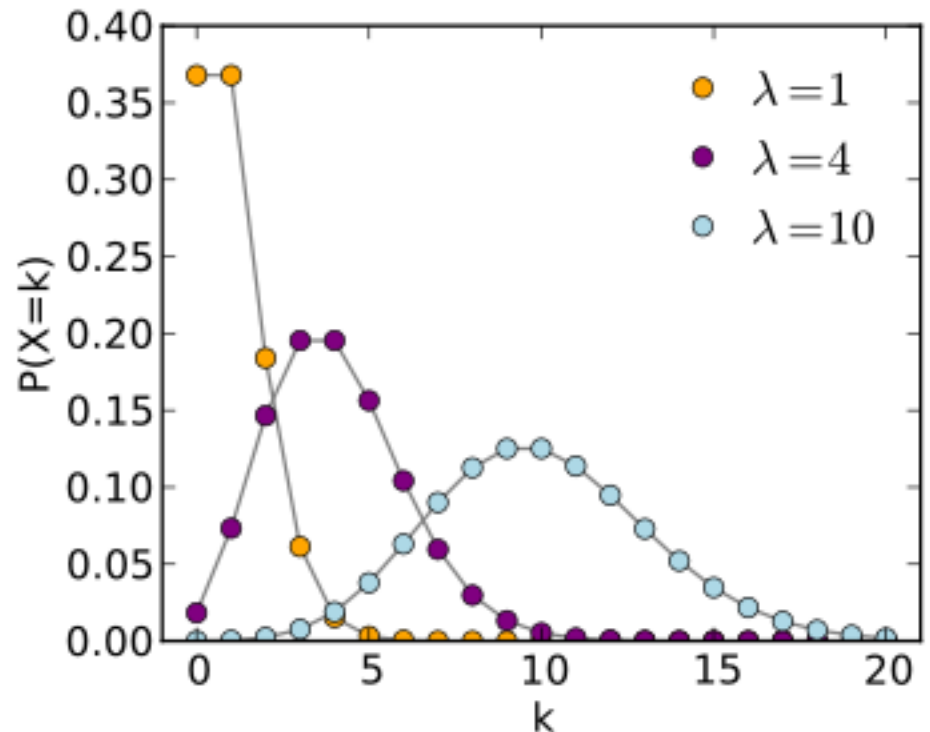
Formulation comes from the limit of the binomial equation

Resembles a normal distribution, but over the positive values, and with only a single parameter.

## **Key properties:**

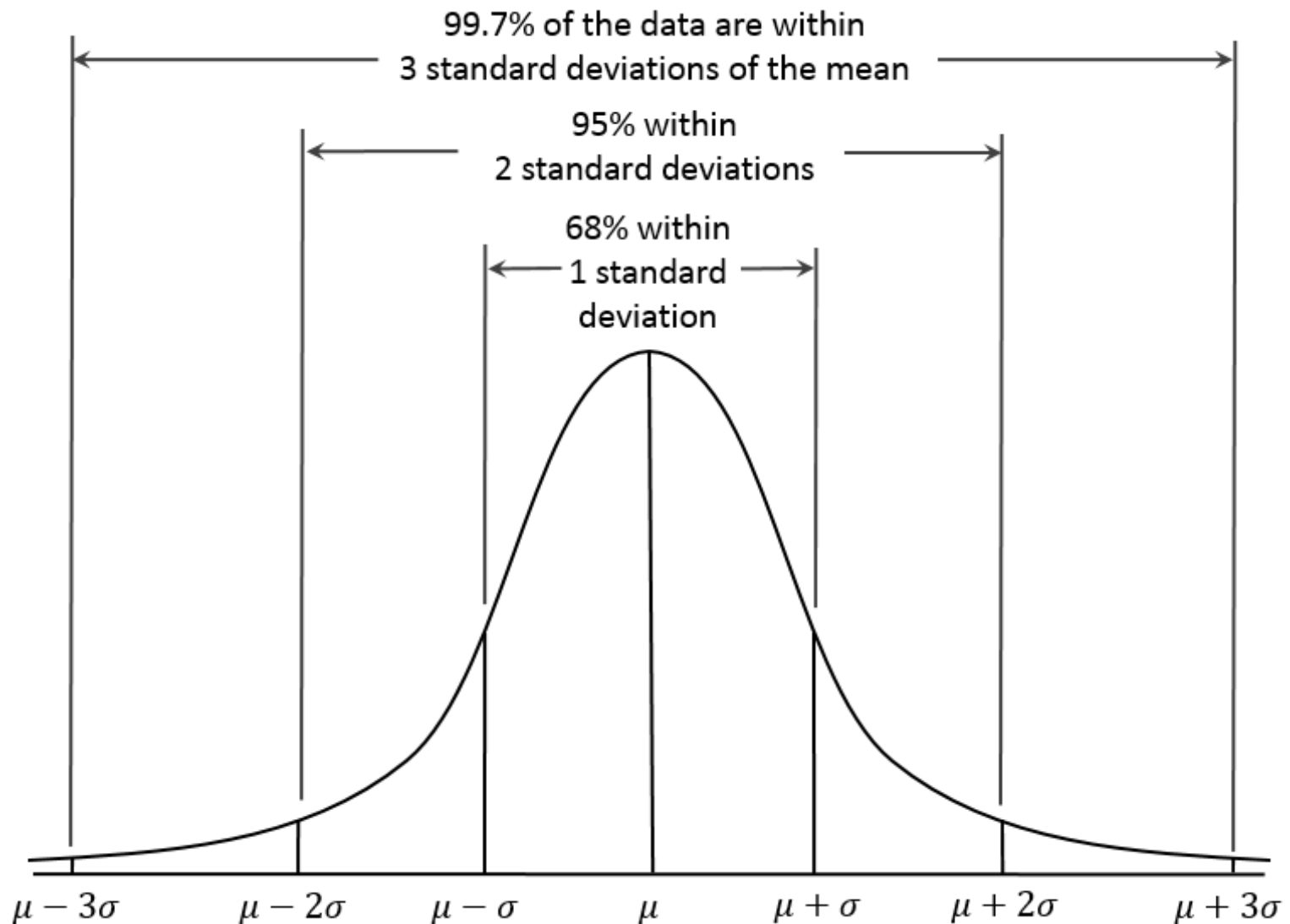
- ***The standard deviation is the square root of the mean.***
- ***For mean > 5, well approximated by a normal distribution***

$$P(k) = \frac{\lambda^k}{k!} e^{-\lambda}$$





# Normal Approximation



# Pop Quiz!

1Mbp = 1 million bases

I want to sequence a 10Mbp genome to 24x coverage.  
How many 120bp reads do I need?

I need  $10\text{Mbp} \times 24x = 240\text{Mbp}$  of data  
 $240\text{Mbp} / 120\text{bp} / \text{read} = 2\text{M}$  reads

I want to sequence a 10Mbp genome so that  
>97.5% of the genome has at least 24x coverage.  
How many 120bp reads do I need?

Find X such that  $X - 2 \times \sqrt{X} = 24$

$$36 - 2 \times \sqrt{36} = 24$$

I need  $10\text{Mbp} \times 36x = 360\text{Mbp}$  of data  
 $360\text{Mbp} / 120\text{bp} / \text{read} = 3\text{M}$  reads

# K-mers and K-mer Counting

# K-mers and K-mer counting

GATTACATACACATTGGATG

# K-mers and K-mer counting

GATTACATACACATTGGATG

GAT ACA ACA ATT GAT

ATT CAT CAC TTG ATG

TTA ATA ACA TGG

TAC TAC CAT GGA

## Kmers:

- Divide a string into substrings of length  $k$
- Notice every position is covered  $k$  times
- Notice there are  $G - k + 1$  kmers from a string of length  $G$

# K-mers and K-mer counting

GATTACATACACATTGGATG

GAT ACA ACA ATT GAT

ATT CAT CAC TTG ATG

TTA ATA ACA TGG

TAC TAC CAT GGA

## Kmers:

- Divide a string into substrings of length  $k$
- Notice every position is covered  $k$  times
- Notice there are  $G - k + 1$  kmers from a string of length  $G$

**Computation:** Very easy to compute, exact matches, represent 32mers in 64 bits

**Biological:** The “atomic unit” of a sequence, creates a fingerprint of a genome/read

# K-mers and K-mer counting

GATTACATACACATTGGATG

GAT ACA ACA ATT GAT

ATT CAT CAC TTG ATG

TTA ATA ACA TGG

TAC TAC CAT GGA

GAT : 2 CAT : 2 ATG : 1 TGG : 1

ACA : 3 CAC : 1 TTA : 1 TAC : 2

ATT : 2 TTG : 1 ATA : 1 GGA : 1

# K-mers and K-mer counting

**GATTACATACACATTGGATG**

**GAT : 2   CAT : 2   ATG : 1   TGG : 1**

**ACA : 3   CAC : 1   TTA : 1   TAC : 2**

**ATT : 2   TTG : 1   ATA : 1   GGA : 1**

**1 : 7   (ATG , TGG , ...)**

**2 : 4   (GAT , CAT , ATT , TAC)**

**3 : 1   (ACA)**



# K-mers and K-mer counting

**GATTACATACACATTGGATG**

**1: 7 (ATG, TGG, ...)**

**2: 4 (GAT, CAT, ATT, TAC)**

**3: 1 (ACA)**

How long should k be?

# K-mers and K-mer counting

**GATTACATACACATTGGATG**

- 1 : 7 (ATG, TGG, ...)**
- 2 : 4 (GAT, CAT, ATT, TAC)**
- 3 : 1 (ACA)**

**How long should k be?**

K=1 : Too short, every base is present

K=2 : Too short, every pair of bases will be present

Pick k so that  $G/(4^k) \ll 1$

$k = \log_4 (G)$

At least 15 for human, often a bit longer

But not too long or could lose resolution

# Coverage Statistics

$$\text{sequencing\_coverage} = \frac{\text{total\_bases\_sequenced}}{\text{genome\_size}}$$

$$\text{genome\_size} = \frac{\text{total\_bases\_sequenced}}{\text{sequencing\_coverage}}$$

$$\text{genome\_size} = \frac{100\text{Gb}}{50\text{x}} = 2\text{Gb}$$

But how can you figure out  
the coverage without a genome?

# K-mer counting

## Kmer-ize

Read 1: GATTACA => GAT, ATT, TTA, TAC, ACA  
Read 2: TACAGAG => TAC, ACA, CAG, AGA, GAG  
Read 3: TTACAGA => TTA, TAC, ACA, CAG, AGA

list

|     |     |        |
|-----|-----|--------|
| GAT | ACA | ACA: 3 |
| ATT | ACA |        |
| TTA | ACA |        |
| TAC | AGA | AGA: 2 |
| ACA | AGA |        |
| TAC | ATT | ATT: 1 |
| ACA | CAG | CAG: 2 |
| CAG | CAG |        |
| AGA | GAG | GAG: 1 |
| GAG | GAT | GAT: 1 |
| TTA | TAC | TAC: 3 |
| TAC | TAC |        |
| ACA | TAC |        |
| CAG | TTA | TTA: 2 |
| AGA | TTA |        |

sort count

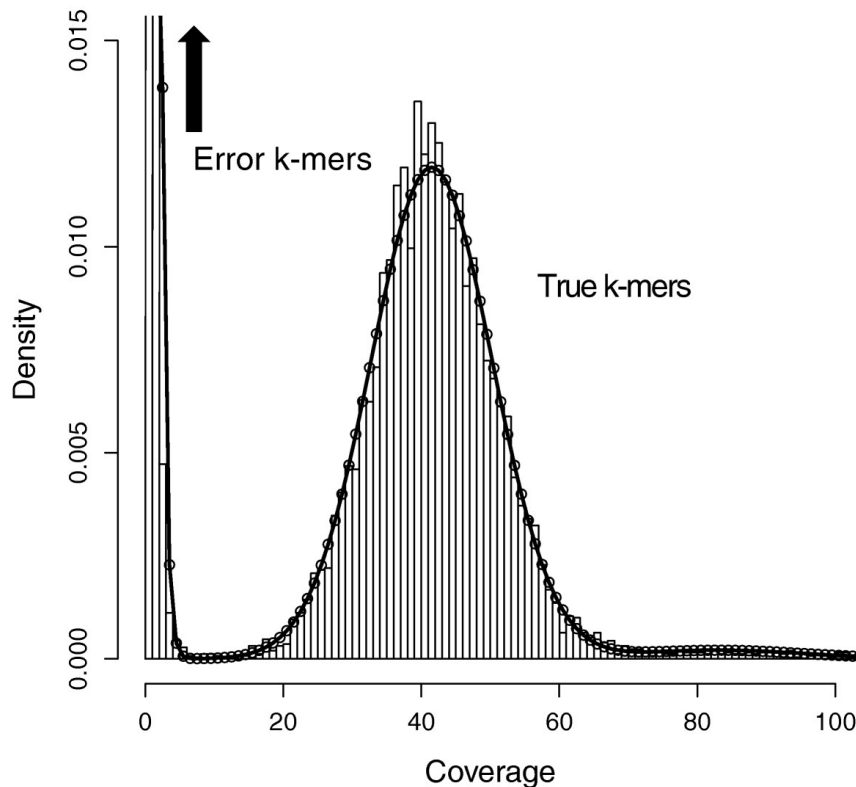
3 kmers occur 1x  
3 kmers occur 2x  
2 kmers occur 3x

tally

From read k-mers alone, can learn something about how frequently different sequences occur (aka coverage)

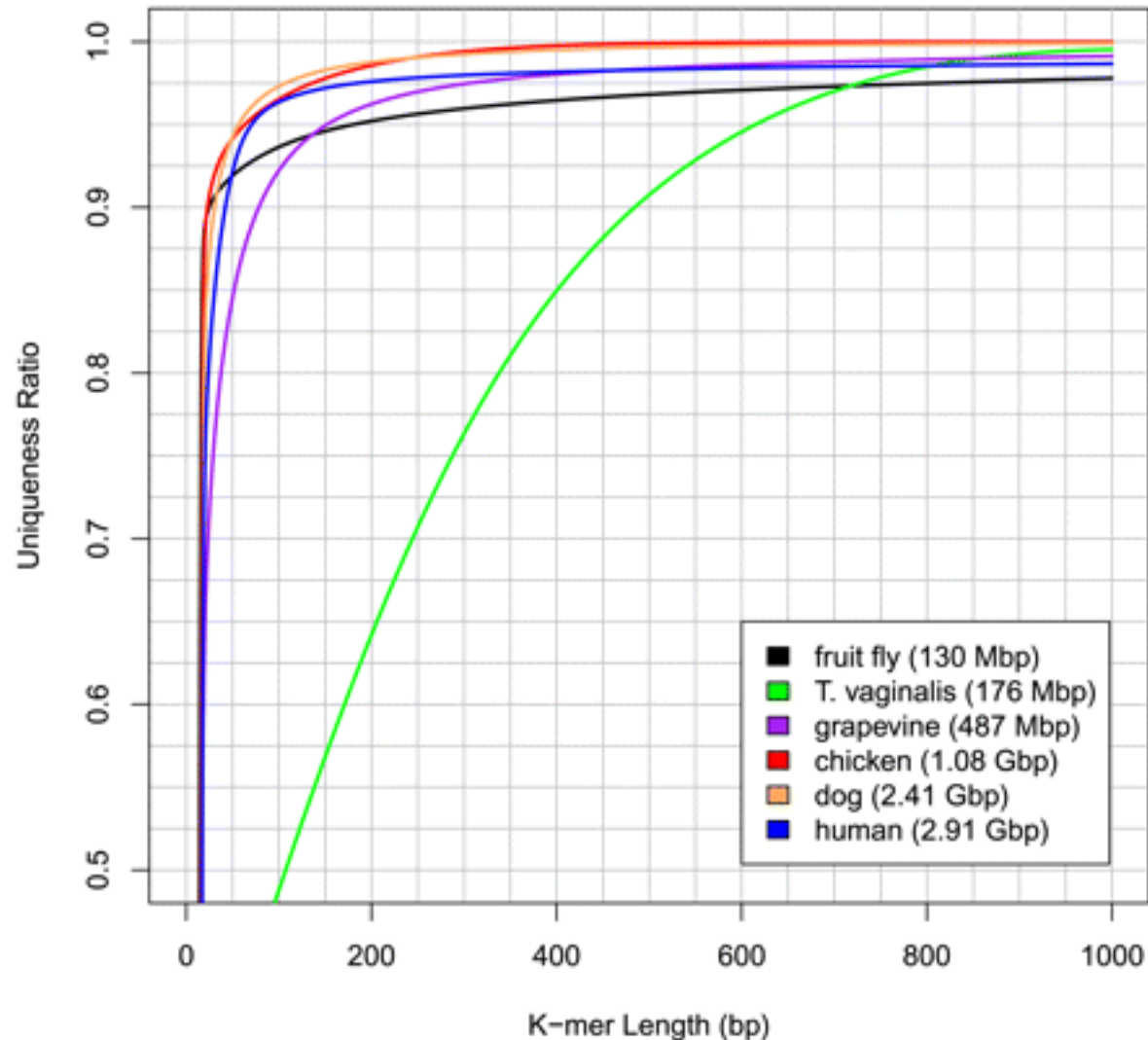
Fast to compute even over huge datasets

# K-mer counting in real genomes



- The tally of k-mer counts in real genomes reveals the coverage distribution.
- Here we sequenced 120Gb of reads from a female human (haploid human genome size is 3Gb), and indeed we see a clear peak centered at 40x coverage
- There are also many kmers that only occur  $<5$  times. These are from errors in the reads
- There are also kmers that occur many times ( $>>70$  times). These are repeats in the genome

# K-mer Uniqueness



**Assembly of large genomes using second-generation sequencing**

Schatz et al. (2010) Genome Research. doi: 10.1101/gr.101360.109

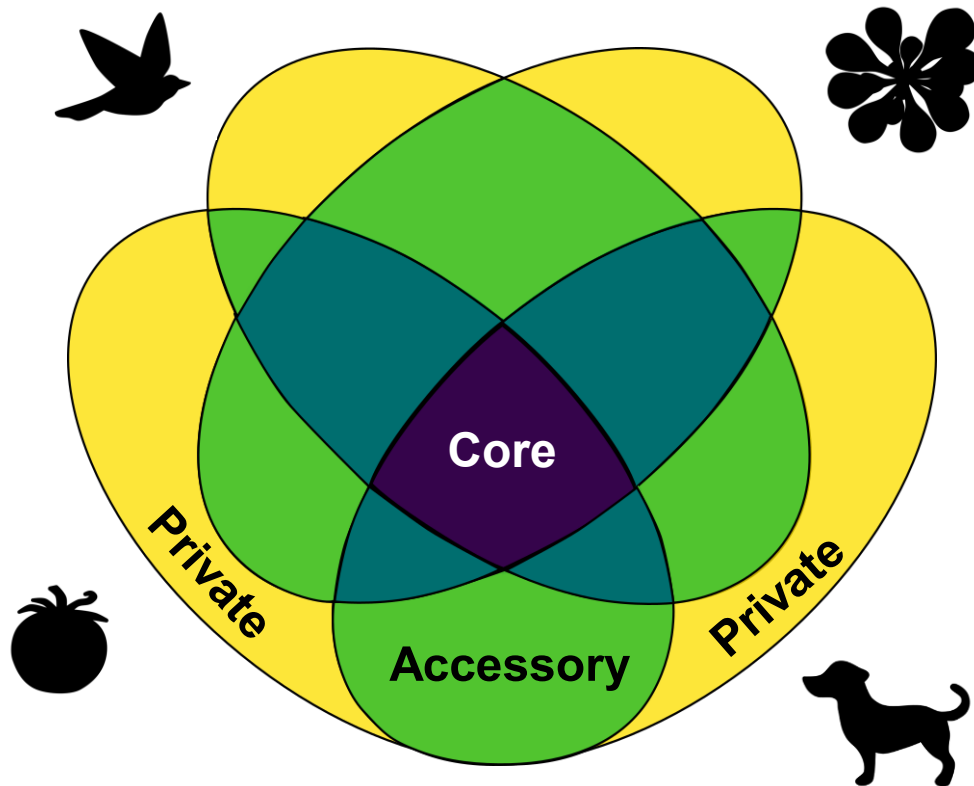
Panagram!

# Pangenomes allow us to study traits across organisms





# Analyzing pangenomes is challenging



Thanks to optimized assembly algorithms, high quality genomes are becoming available for many samples

- How to compare them?
- How to find sequences that are shared or private across samples?
- How to relate to traits?
- How to do this at scale?

# K-mer based analysis

Katharine  
Jenike



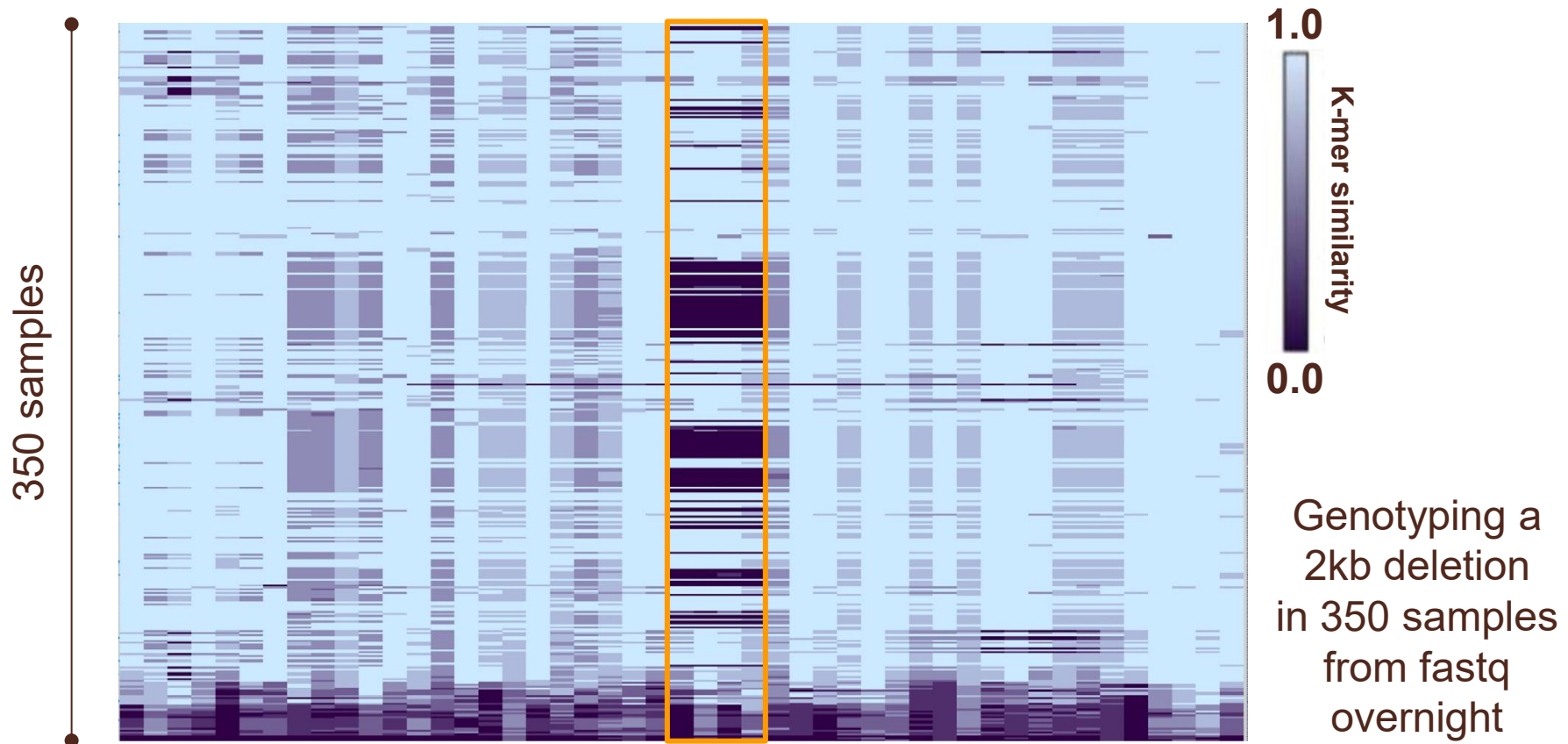
|       |                                            |   |
|-------|--------------------------------------------|---|
| equal | ...GATCCGTAGCTTACGGAATCC...                | 1 |
|       | ...GATCCGTAGCTTACGGAATCC...                | 2 |
|       | ...111111111111111111111111...             |   |
| sub   | ...GATCCGTAGCTTACGGAATCC...                | 1 |
|       | ...GATCCGTAGC <b>GC</b> ATACGGAATCC...     | 2 |
|       | ...11111111 <b>00000</b> 11111111...       |   |
| del   | ...GATCCGTAGCTTACGGAATCC...                | 1 |
|       | ...GATCCGTAG <b>---</b> ACGGAATCC...       | 2 |
|       | ...1111111 <b>0000000</b> 11111111...      |   |
| ins   | ...GATCCGTAG      ACGGAATCC...             | 1 |
|       | ...GATCCGTAG <b>CTT</b> ACGGAATCC...       | 2 |
|       | ...1111111 <b>00</b> <b>00</b> 11111111... |   |

- Extremely fast
- Memory efficient (one-hot encoding)
- Captures all types of mutations
- Each mutation impacts at least k bases
  - k=3 here

## k-mer approaches for biodiversity genomics

Jenike *et al.* (2024) *Genome Research*. doi: 10.1101/gr.279452.124

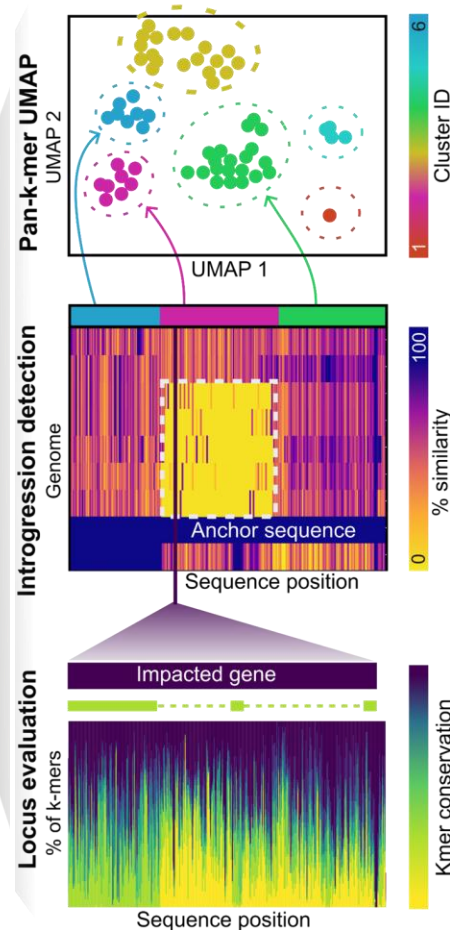
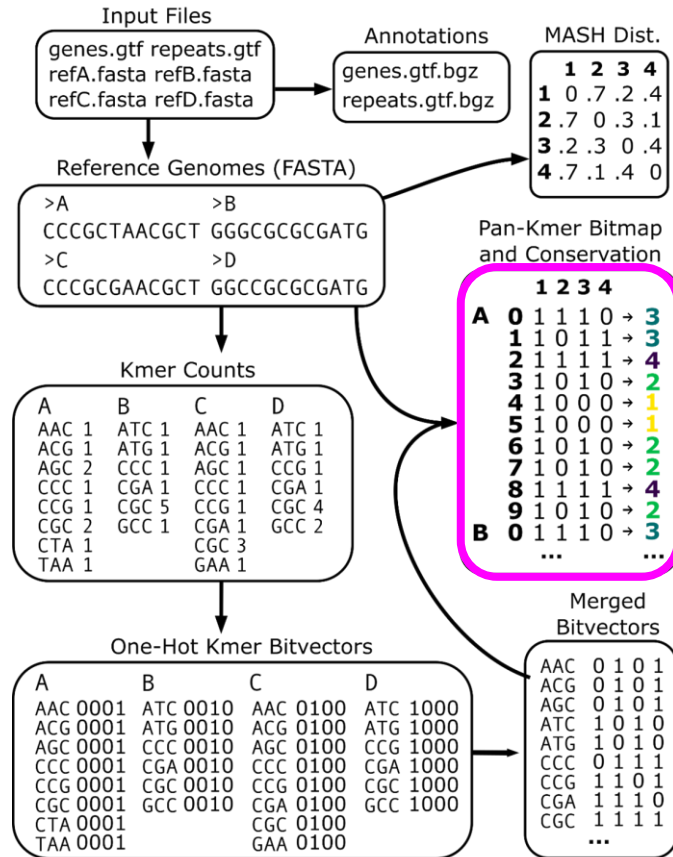
# K-mer based genotyping



**A neofunctionalized flowering antagonist created an evolutionary contingency that channeled Solanaceae adaptation**

Shohat *et al.* (2026) *bioRxiv*. doi: 10.64898/2026.02.20.707063

# Panagram's K-mer bitmap



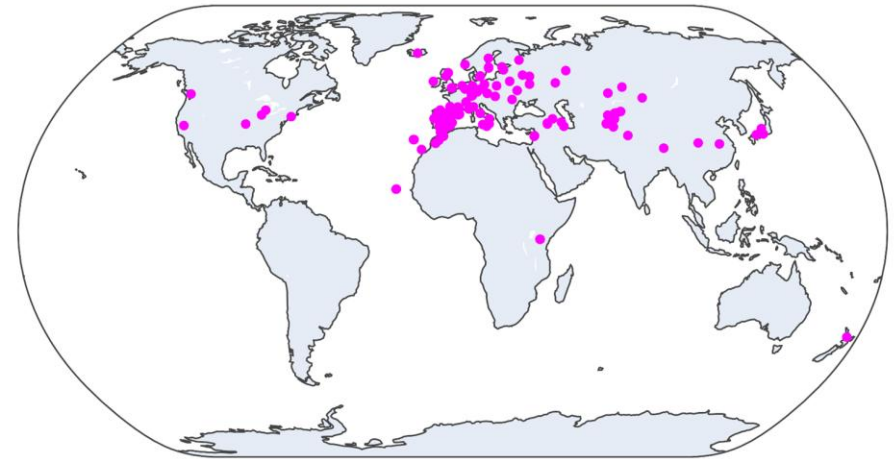
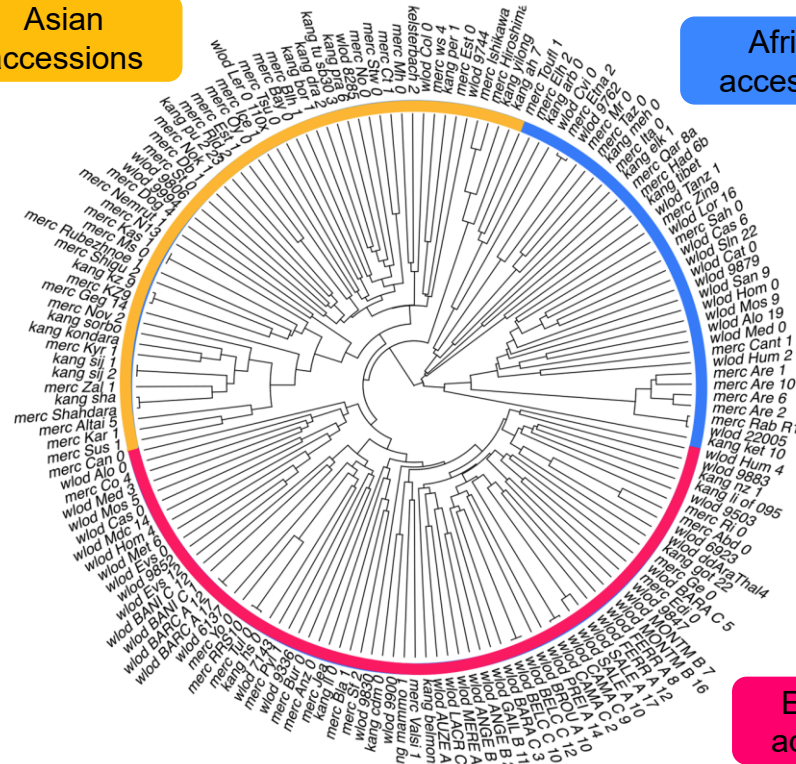
The panagram bitmap includes the **presence/absence of every k-mer** in the pan-genome.

Interesting regions can be found by looking at where k-mers are very conserved or very unique across the pangenome

# Globally diverse *A. thaliana* genomes

Asian  
accessions

African  
accessions



European  
accessions

160 genomes took 1hr 12  
minutes with k=51



**Robin Burns**

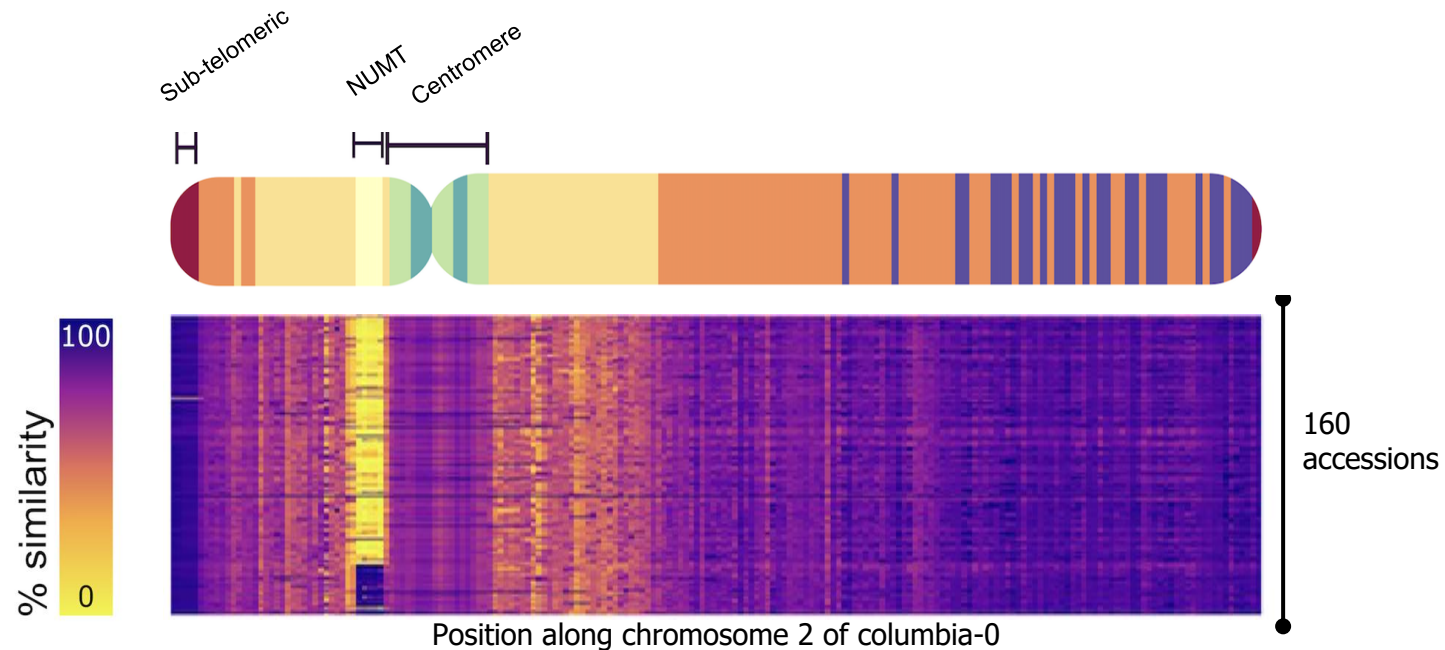
Assemblies from:

Wlodzimierz et al. Nature 2023, Lian et al. Nature Genetics 2024, Kang et al. Nat. Com. 2023





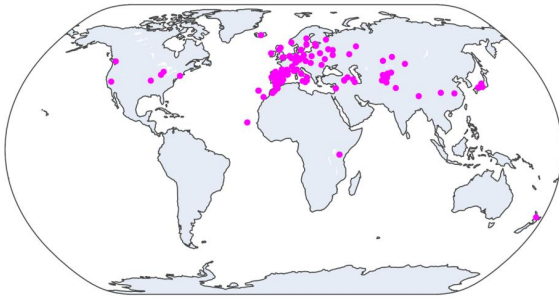
# Annotating *A. thaliana* pangenome



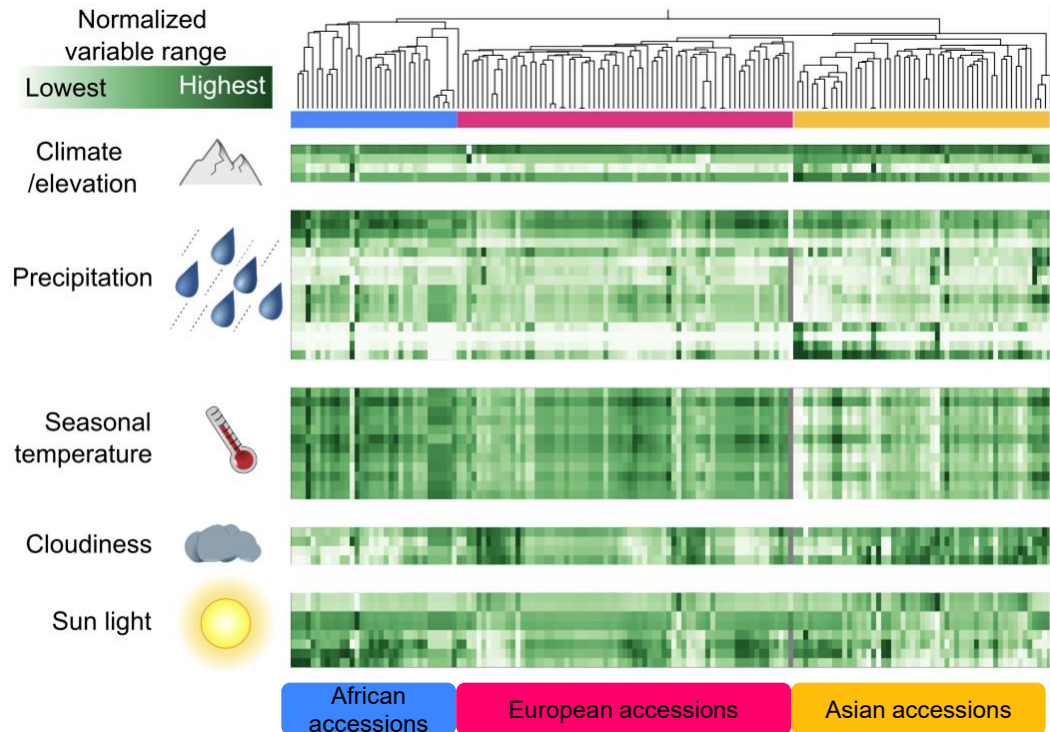
Dark purple = more similar to reference accession

Lighter = less similar to reference accession

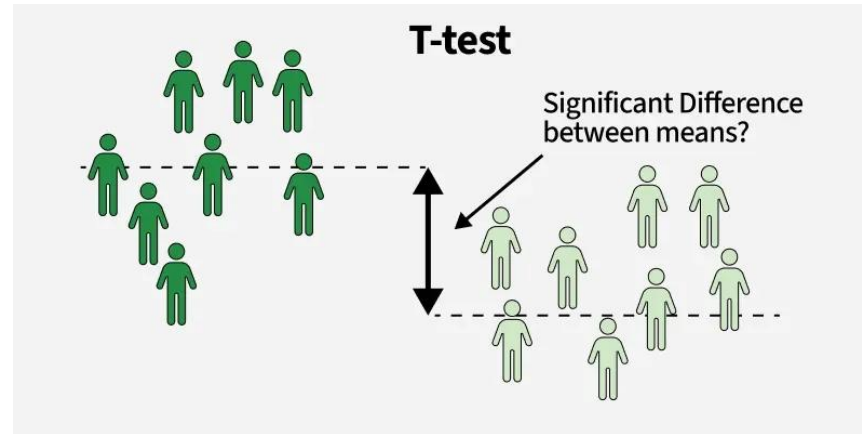
# *A. thaliana*'s environment



How do you check if an environmental condition is associated with a change in the genome?

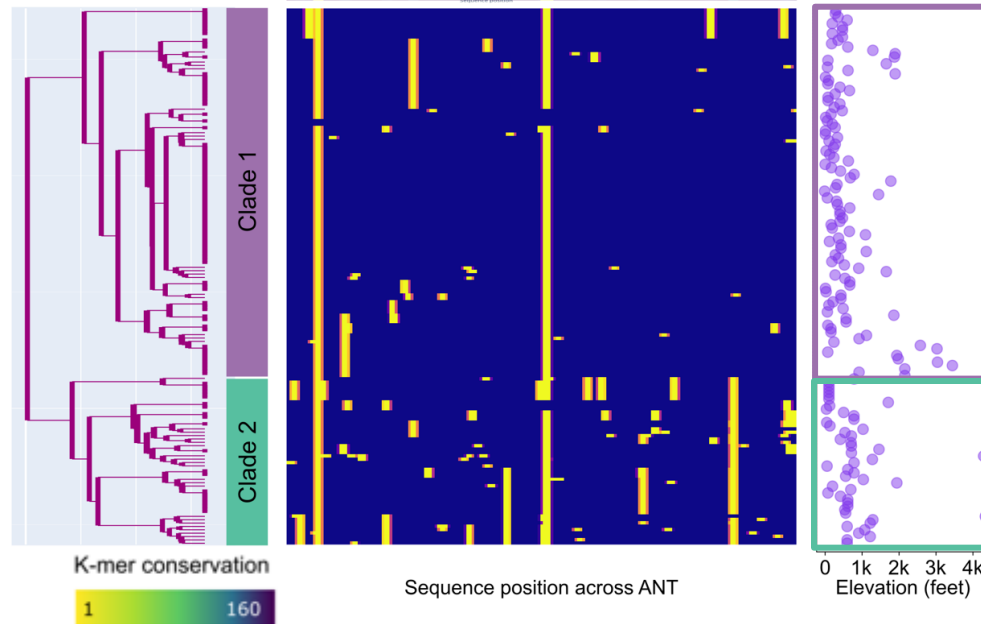
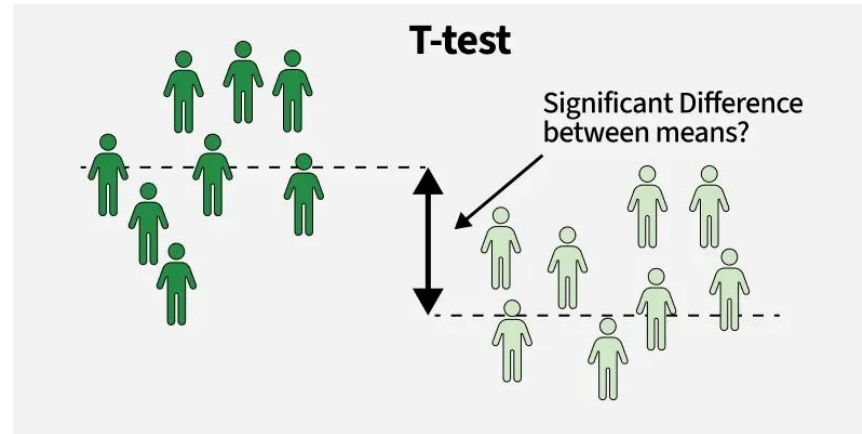


# Association testing

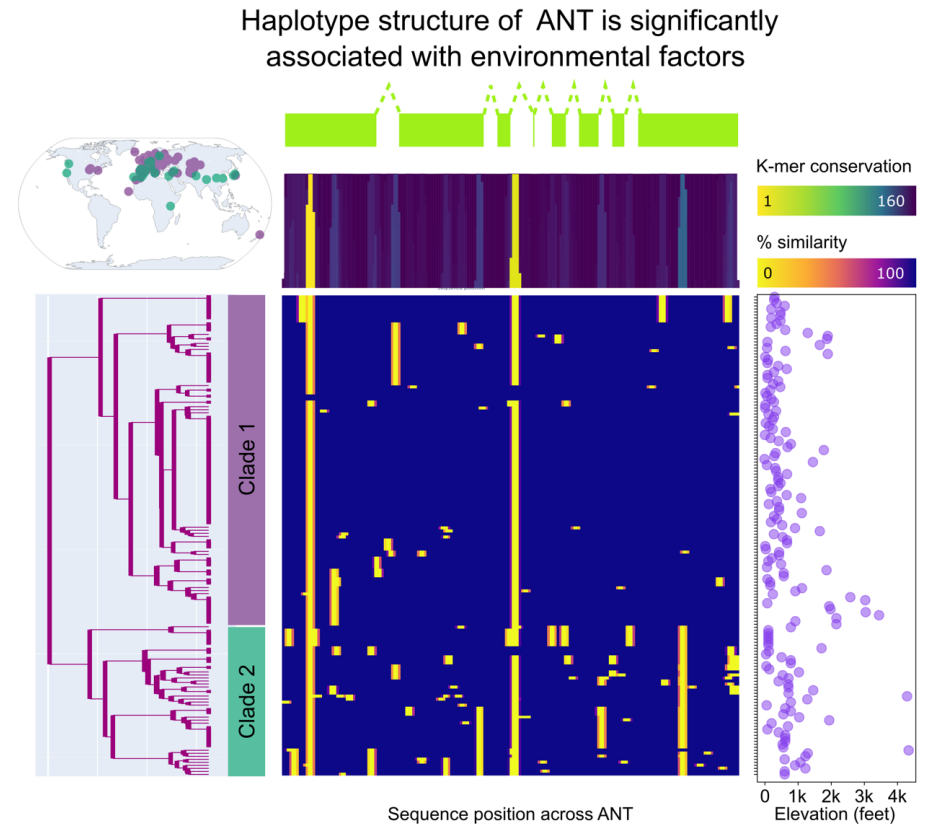
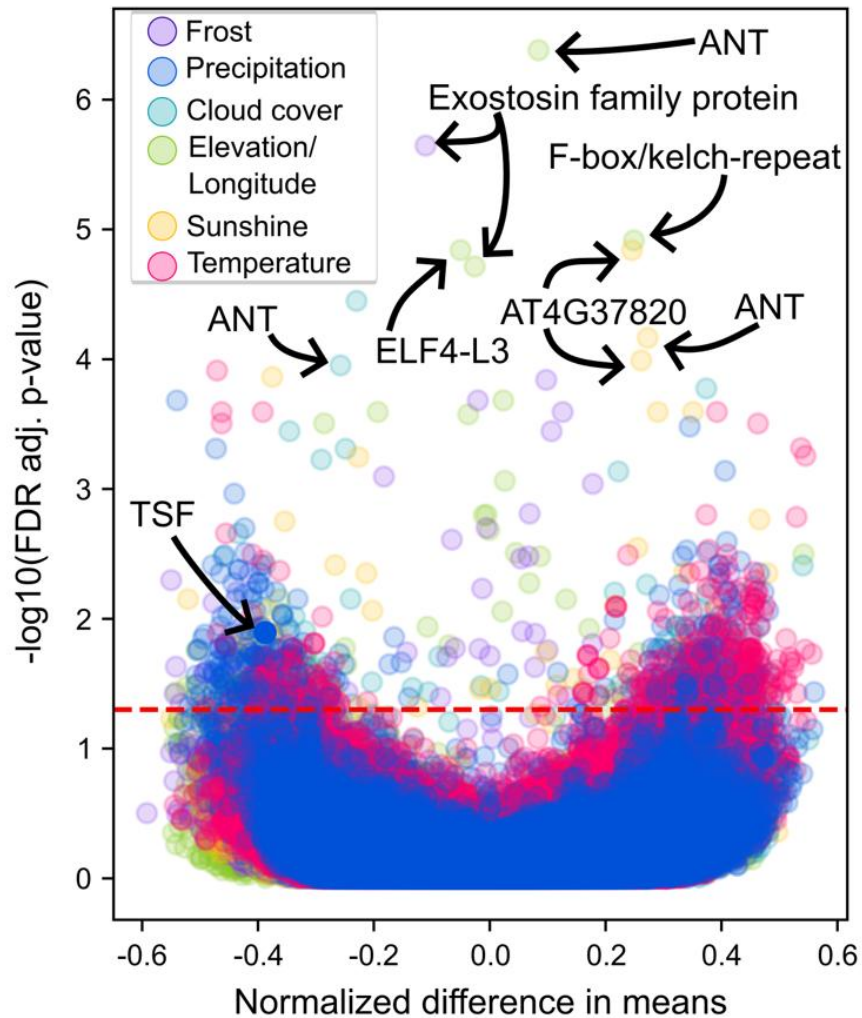




# Association testing



# Association testing in *A. thaliana*



# Association testing in *Canis*



**Appa (M)**  
Sheepadoodle



**Molly (F)**  
Greenland Village



**Orfhlaith (F)**  
American Mutt



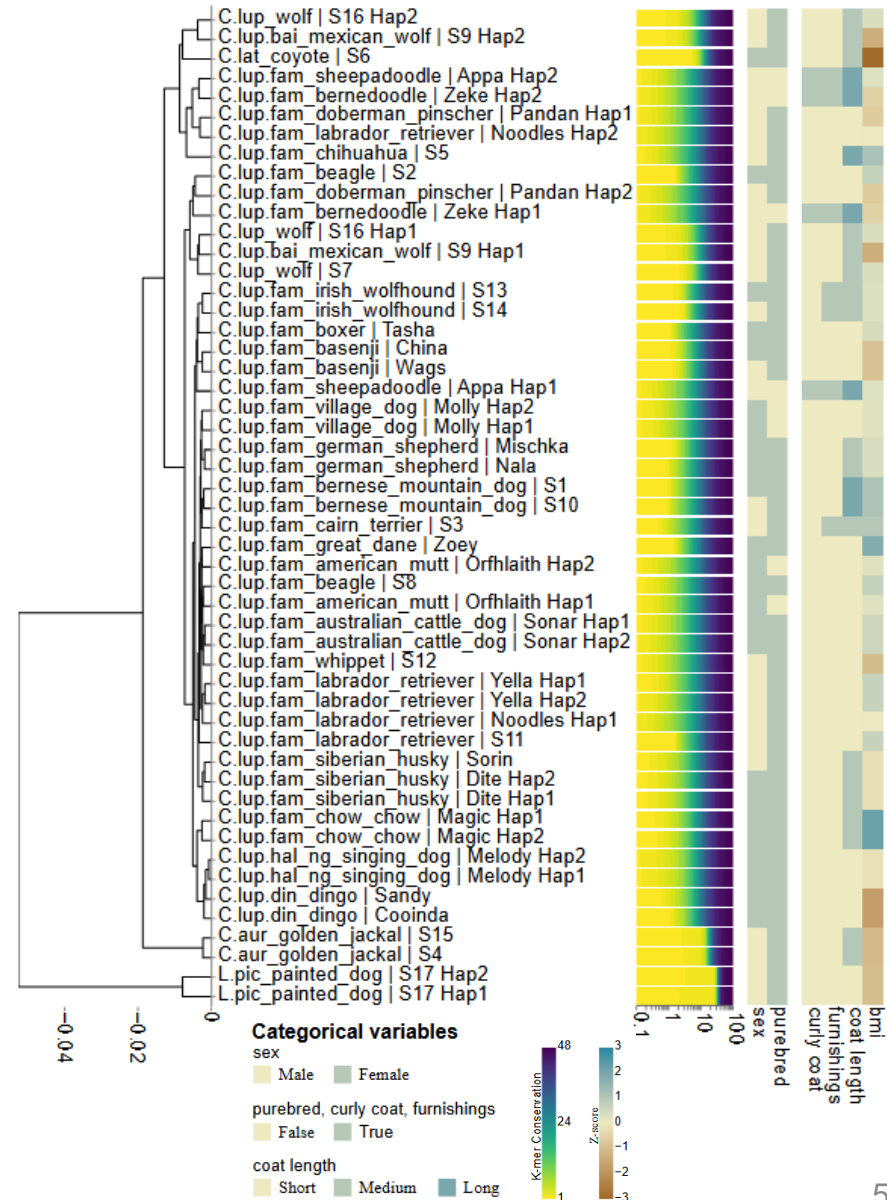
**Zeke (M)**  
Bernedoodle



**Noodle (M)**  
Labrador retriever

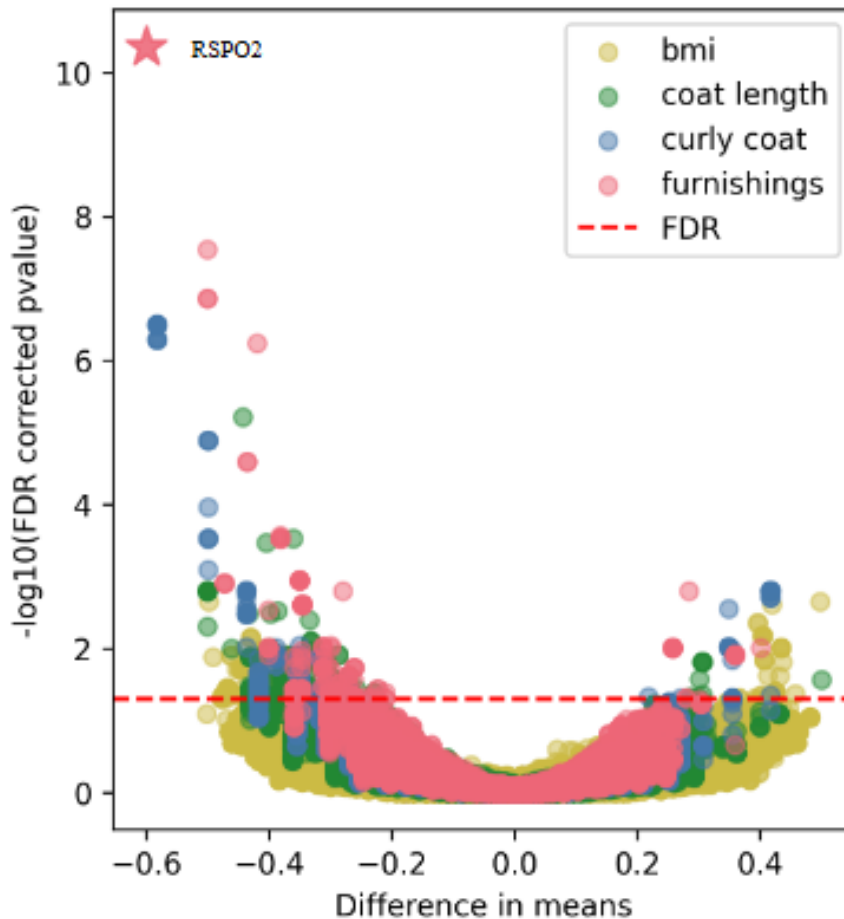


**Pandan (M)**  
Doberman

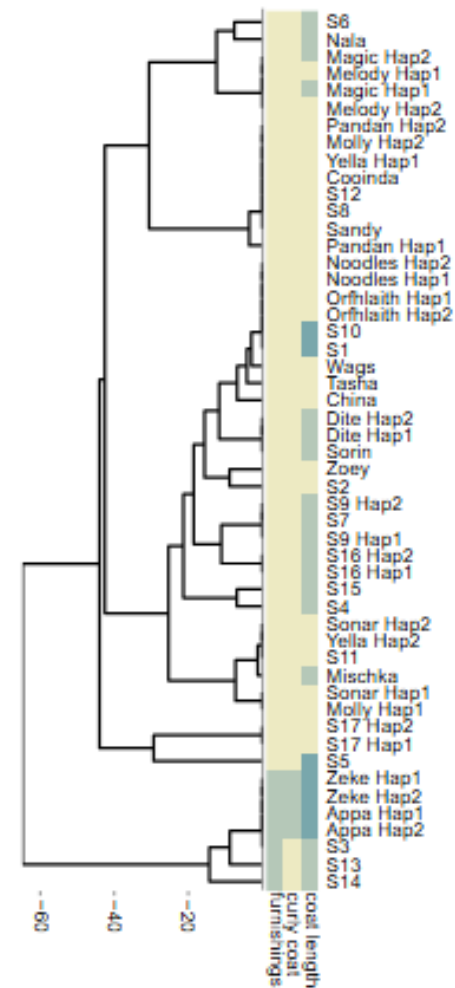


# Association testing in *Canis*

Associations between exons and phenotypes



RSP02 exon structure clustering



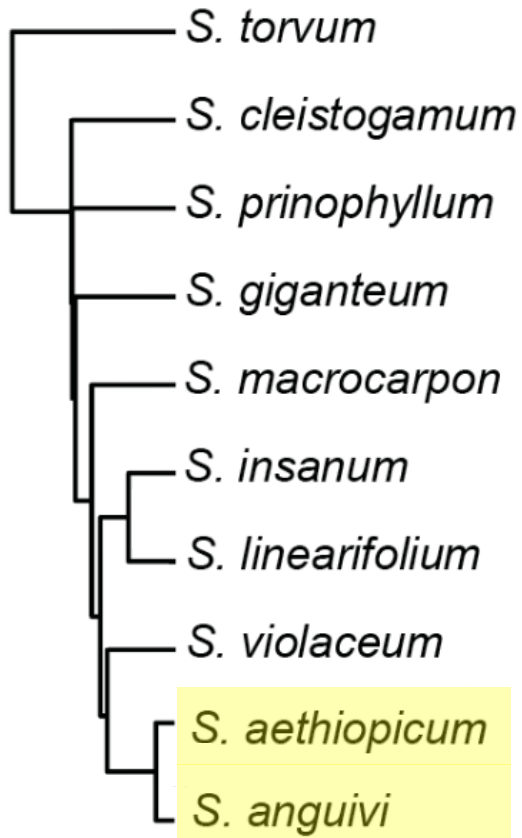


# Panagram can identify introgressions



- **Introgressions:** chromosomal regions from one species that are introduced into the gene pool of another through hybridization and backcrossing
- Some Impacts Include:
  - **Resistance** to insect damage, diseases, and extreme heat/cold
  - **Variations** in flower/fruit shape, color, and texture in plants
  - **Modifications** of appearance or behavior in animals

# Panagram can identify introgressions



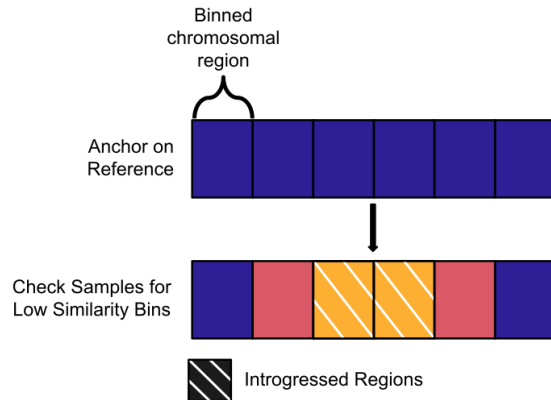
Aethiopicum  
(Domesticated)



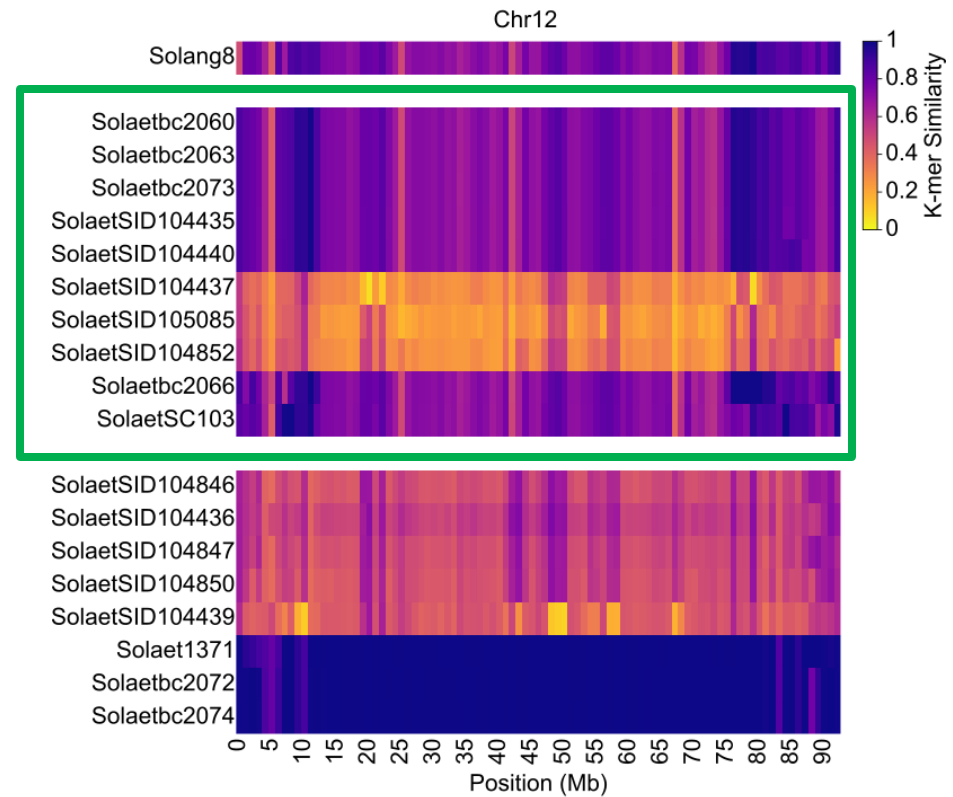
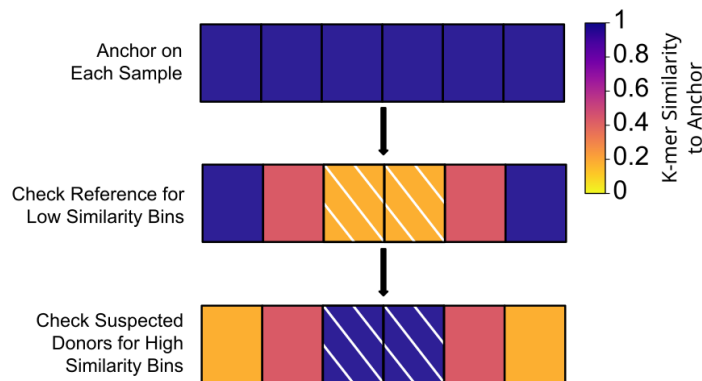
Anguivi  
(Wild)

# Panagram can identify introgressions

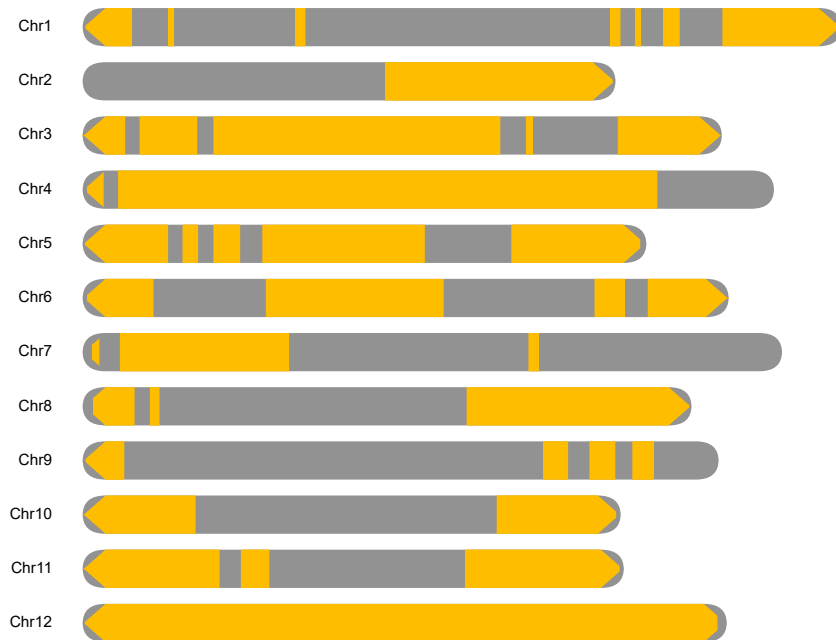
## 2-way introgression approximation



## 3-way introgression calling



# Annotating found introgressions



- Aethiopicum introgressions impact:
  - Ribosomal and Translational Proteins
  - Transcription Factors and DNA Binding Proteins
  - Response to Environment
    - TAS14
    - SP5G/SP3D
    - HSP21
  - Protein Modification and Degradation



# Illumina Sequencing Summary

## Advantages:

- Best throughput, accuracy and read length for any 2nd gen. sequencer
- Fast & robust library preparation

## Disadvantages:

- Inherent limits to read length (practically, 150bp)
- Some runs are error prone
- Requires amplification, sequences a population of molecules



### Illumina HiSeq

~3 billion paired 100bp reads  
~600Gb, \$10K, 8 days  
(or “rapid run” ~90Gb in 1-2 days)

### Illumina NextSeq

One human genome in **<30 hours**

### Illumina NovaSeq

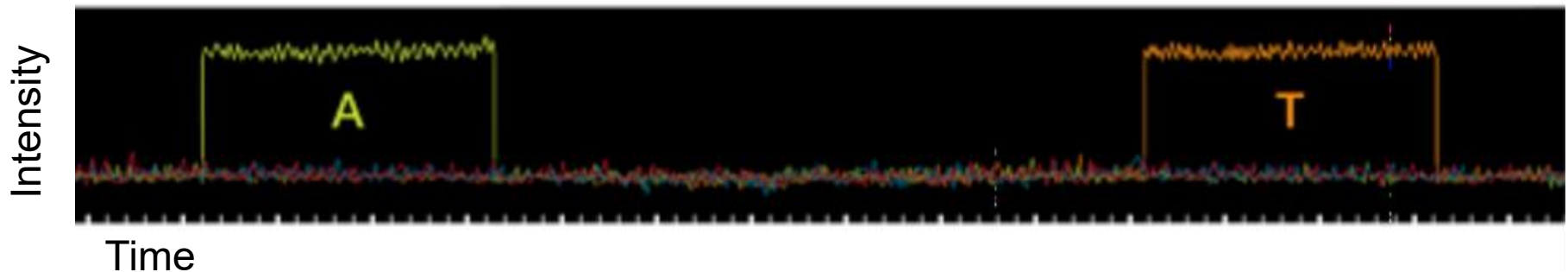
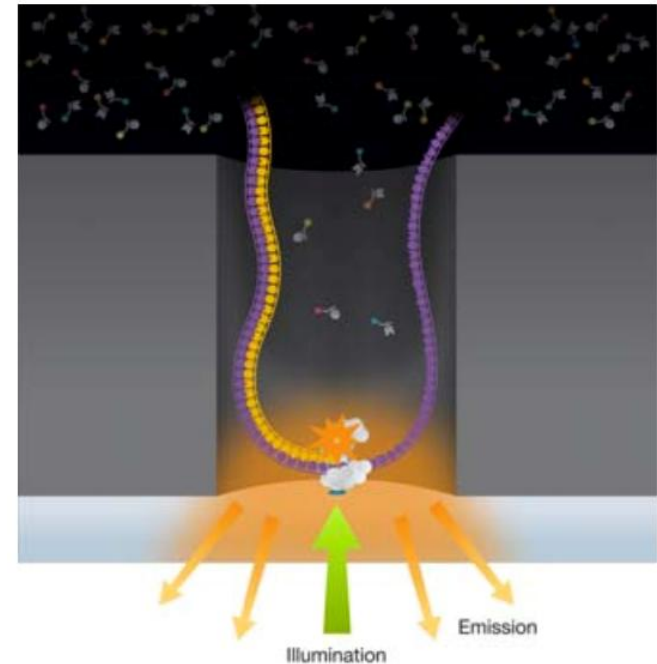
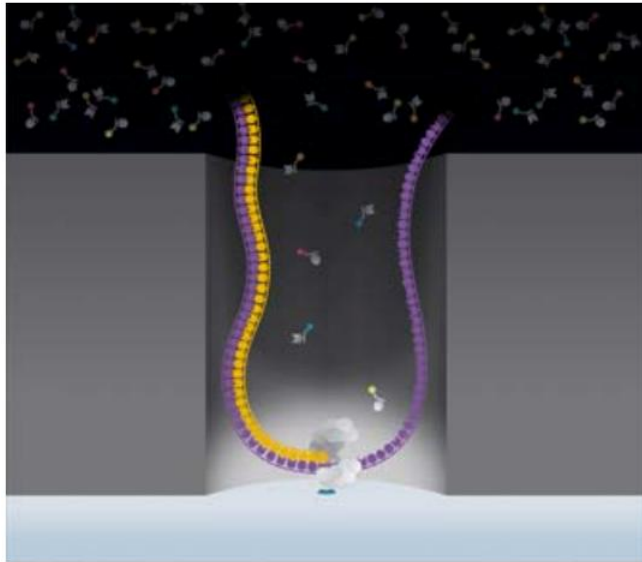
48 Genomes / run  
>8000 genomes per year  
3Tb, <3 days, ~\$300 / genome

# PacBio: SMRT Sequencing



# PacBio: SMRT Sequencing

Imaging of fluorescent phospholinked labeled nucleotides as they are incorporated by a polymerase anchored to a Zero-Mode Waveguide (ZMW).



# “HiFi” Circular Consensus Reads

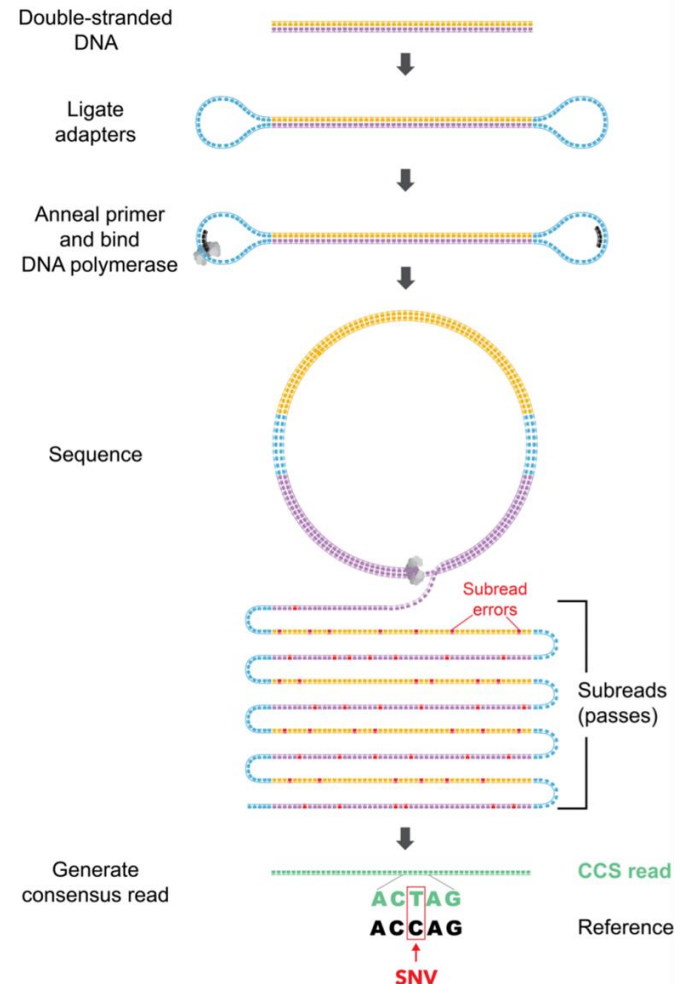
High-quality reads produced by sequencing the same molecule multiple times

Higher accuracy for low-coverage sequences like somatic variants or lowly expressed transcripts in RNA-seq, more interpretable alignments, better & faster assembly

Limits read length, used to be very expensive but more manageable now

**Accurate circular consensus long-read sequencing improves variant detection and assembly of a human genome**

Wenger et al(2019) Nature Biotechnology doi:10.1038/s41587-019-0217-9

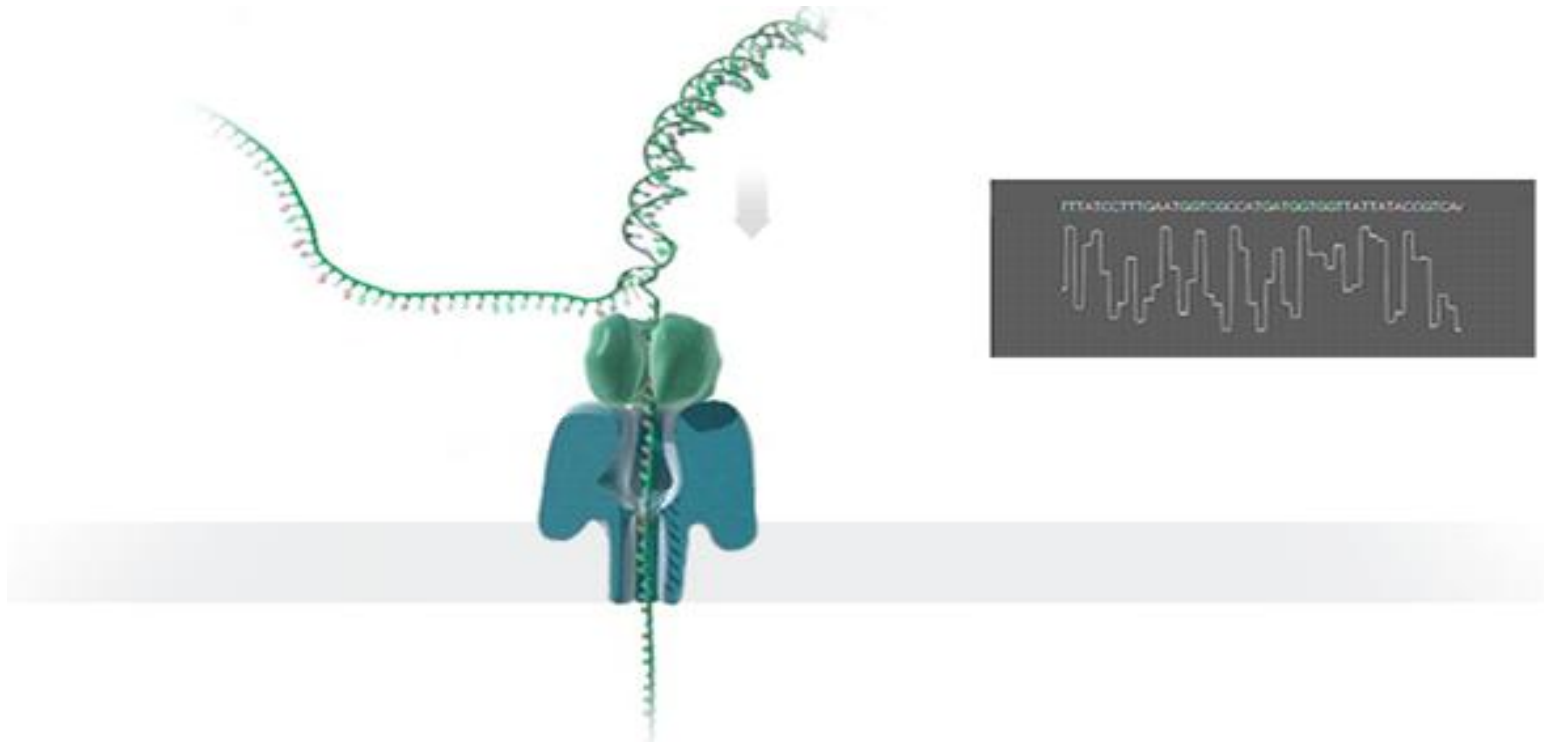


# Oxford Nanopore Technologies (ONT)



# Nanopore Sequencing

Sequences DNA/RNA by measuring changes in ionic current as nucleotide strand passes through a pore

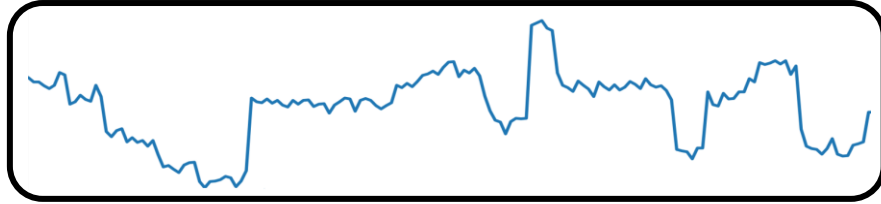


<https://www.youtube.com/watch?v=VxGliKyYuFQ>  
<https://www.youtube.com/watch?v=mCYUxgSEUlc>

[nanoporetech.com/applications/dna-nanopore-sequencing](https://nanoporetech.com/applications/dna-nanopore-sequencing)

# Nanopore Basecalling

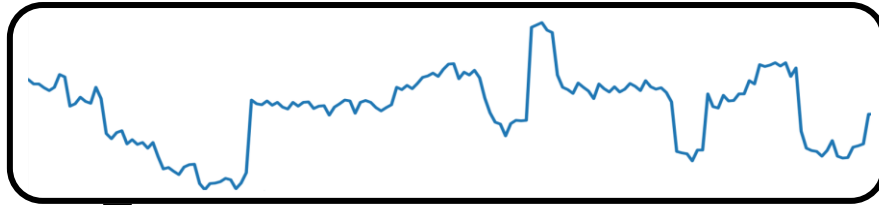
Raw Signal



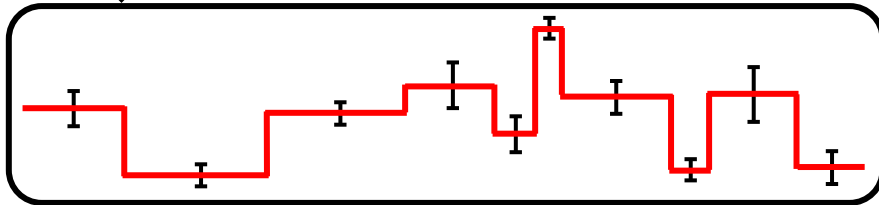
Translation of raw signal  
into basepairs

# Nanopore Basecalling

Raw Signal



Events



Translation of raw signal into basepairs

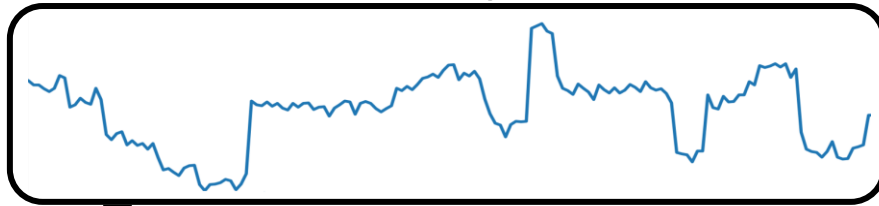
Early basecallers began by estimating k-mer boundaries using “events”, which were then input to an HMM

Modern basecallers use neural networks directly on raw signal

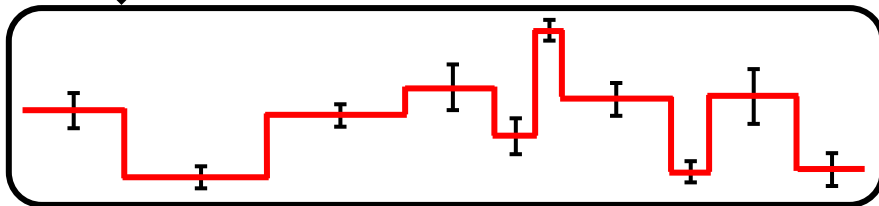


# Nanopore Basecalling

Raw Signal



Events

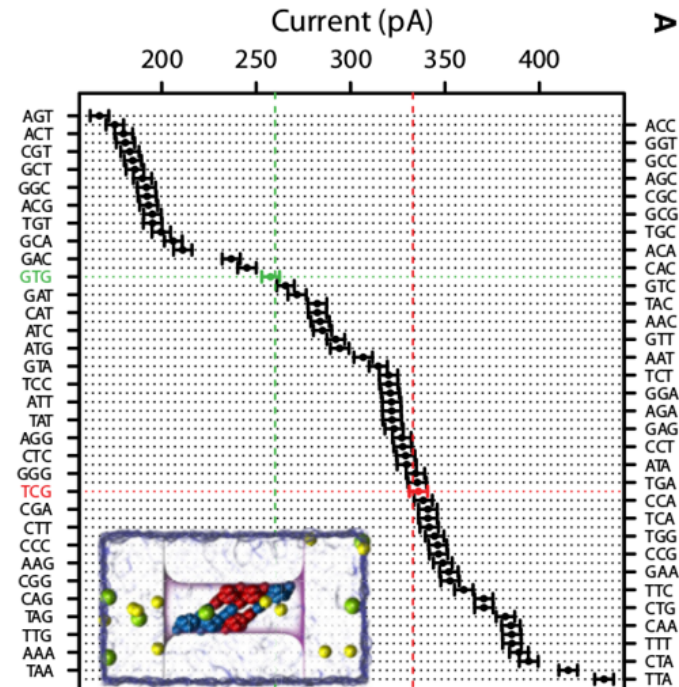


Possible k-mers

| 0    | 1    | 2    | 3    |
|------|------|------|------|
| TCCA | CCAT | CATG | TACA |
| AGCA | TGGC | TTAC | TCCA |
| GTCT | ATTA | ACGT | GACG |
| GATT | ATTG | GTCT | ACGG |

(Based on probability of event matches)

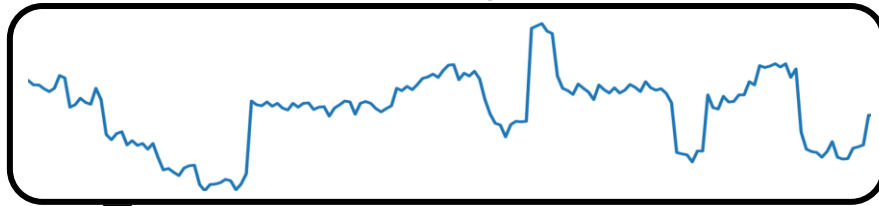
ONT releases k-mer models with expected current distribution of every k-mer



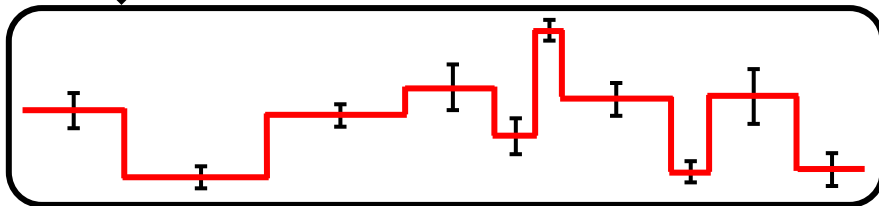
DNA Base-Calling from a Nanopore Using a Viterbi Algorithm  
Timp et al. (2012) *Biophysical Journal*

# Nanopore Basecalling

Raw Signal



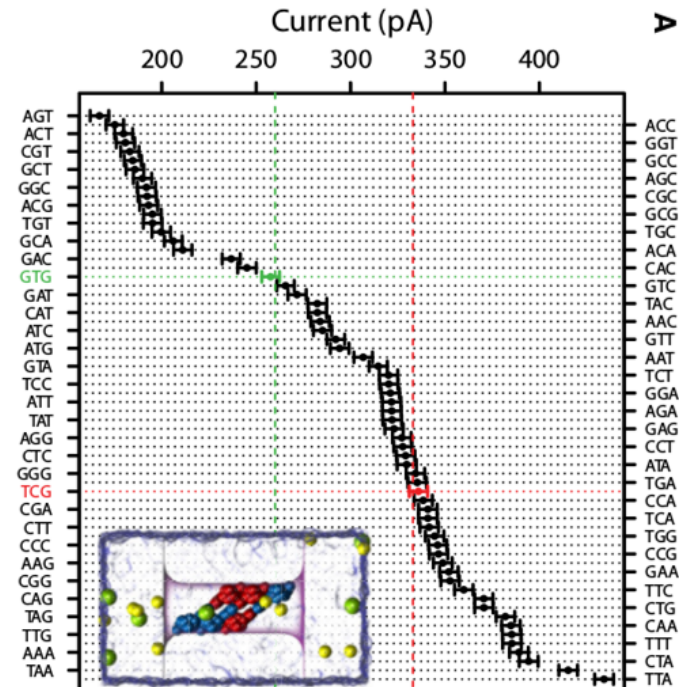
Events



Possible k-mers

| 0               | 1               | 2    | 3    |
|-----------------|-----------------|------|------|
| TCCA → CCAT     | CATG            | TACA |      |
| <del>AGCA</del> | <del>TGGC</del> | TTAC | TCCA |
| <del>GTCT</del> | ATTA            | ACGT | GACG |
| GATT ↗ ATTG     | GTCT            | ACGG |      |

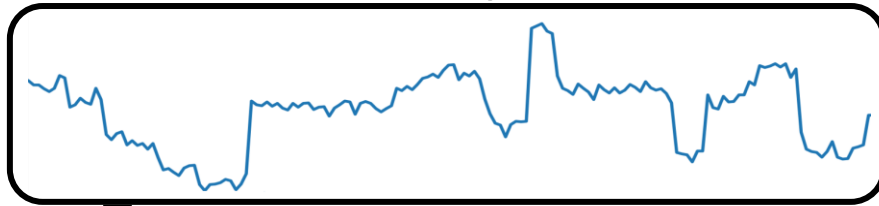
Certain k-mers can be eliminated based on possible transitions



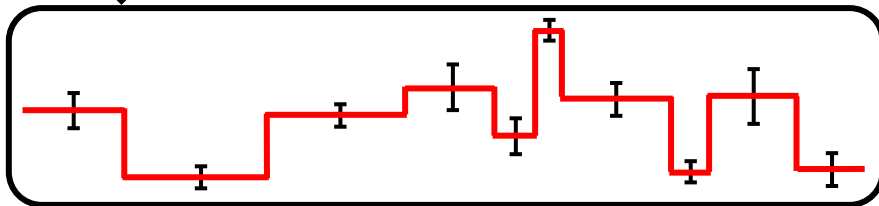
DNA Base-Calling from a Nanopore Using a Viterbi Algorithm  
Timp et al. (2012) *Biophysical Journal*

# Nanopore Basecalling

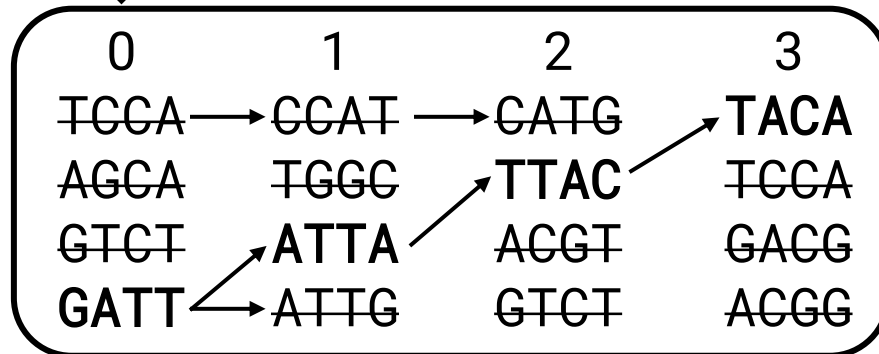
Raw Signal



Events

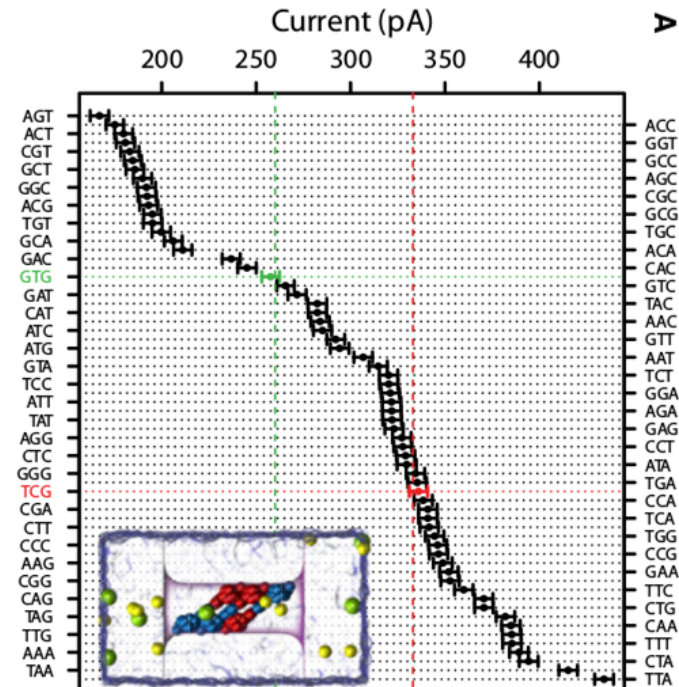


Possible k-mers



**GATTACA**

Final sequence determined by most probable k-mers



“DNA Base-Calling from a Nanopore Using a Viterbi Algorithm”  
Timp et al. (2012) *Biophysical Journal*